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# IMPLICATIONS OF PALEOMAGNETIC DATA ON MIOCENE EXTENSION NEAR A MAJOR ACCOMMODATION ZONE IN THE BASIN AND RANGE PROVINCE, NORTHWESTERN ARIZONA AND SOUTHERN NEVADA

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Abstract. Paleomagnetic data from volcanic and crystalline rocks elucidate the evolution of a major Miocene accommodation zone in the northern Colorado River extensional corridor. The accommodation zone is a 10-km-wide belt of intermeshing conjugate normal faults that facilitates reversals in the dominant tilt direction of fault blocks and dip direction of major normal fault systems. Tilt-corrected means (e.g., N = 28 sites, D = 353.4°, I = 61.3°, $\alpha_{95}$= 6.7°, k = 17.4) from Miocene volcanic strata overlap expected Miocene directions at the 95% confidence level. These data and geologic relations suggest that at exposed structural levels the accommodation zone did not facilitate distributed strike-slip displacement between opposing tilt block domains. Vertical axis rotations are probably negligible in most of the corridor, as the Miocene structural grain generally mimics that in the zone. Discrepancies between characteristic remanent magnetizations (ChRM) in crystalline rocks and expected directions are therefore attributed to rotations about horizontal (i.e., tilting or flexing) axes. ChRMs from Cretaceous and Miocene intrusions suggest 50°-90° of tilting of large crystalline terranes on either side of the accommodation zone. The magnitude of tilting inferred from the paleomagnetic data is similar to that of Tertiary strata in nearby fault blocks, implying that these crystalline terranes are parts of highly tilted fault blocks rather than flexed lower plate rocks. Major low-angle normal faults that bound highly tilted parts of these crystalline terranes probably nucleated at steep dips and were rotated to gentle dips by block tilting. Paleomagnetic data indicative of negligible tilting (e.g., Miocene intrusions, northern Black Mountains crystalline terrane, N = 13 sites, D = 359°, I = 55°, $\alpha_{95}$= 9°, k = 24) and geologic relations imply that lower plate rocks may surface in both the east and west tilted domains 35-50 km away from the zone. The trend toward shallower structural levels, with respect to Miocene extension, and tapering of highly extended terrane toward the accommodation zone imply that the magnitude of upper crustal extension decreases toward the zone. Temporal patterns of major extension, especially an apparently continuous northward younging across both the east and west tilted domains, further suggest that the accommodation zone served as a long-lived rupture barrier between conjugate normal fault systems rather than as a short-term boundary between opposing systems that propagated toward and converged at the zone.

## INTRODUCTION

Structural asymmetry is a conspicuous characteristic of rifted continental crust. Regional consistency in both the tilt direction of fault blocks and dip direction of major normal faults is common. Accommodation zones, or tilt block domain boundaries (Figure 1), facilitate reversals in the dominant direction of block tilting and dip direction of normal fault systems [Bosworth, 1985; Bosworth et al., 1986; Rosendahl, 1987; Ebinger, 1989; Faulds et al., 1990]. Major accommodation zones may coordinate reversals in the dip of regionally extensive detachment systems [Bosworth et al., 1986; Lister et al., 1986]. Thus, the style of deformation in the zones has potential for constraining the magnitude and direction of relative movement between opposing detachment terranes [Faulds et al., 1990]. Because deformational style may vary with structural level along a zone (Figure 1), the lower or upper plate (i.e., footwall or hanging wall of a detachment fault) affinity of crustal blocks must be understood to accurately describe the deformational style.

This paper considers the implications of paleomagnetic data on the structural development of a major Miocene accommodation zone in the northern Colorado River extensional corridor of the Basin and Range province, northwestern Arizona and southern Nevada. K/Ar isotopic age determinations on variably tilted volcanic rocks in major growth-fault basins and on many of the crystalline rocks sampled for paleomagnetic studies are also reported (Table 1). Using paleomagnetic data from layered Miocene volcanic rocks, we define the relative contributions of horizontal and vertical axis rotations, definitions that are critical for determining whether the zone facilitated distributed strike-

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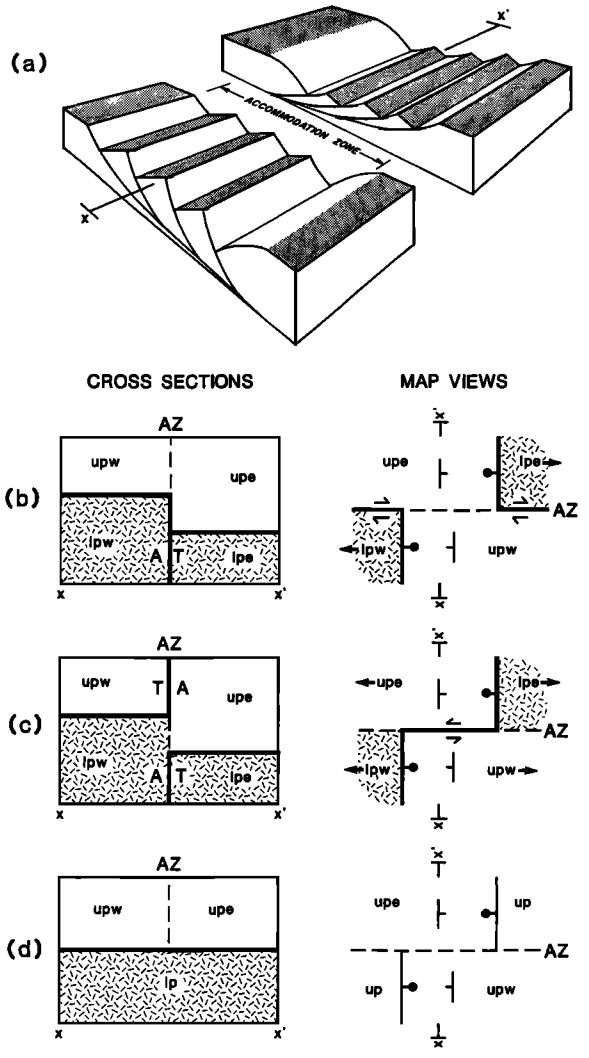



Fig. 1. Schematic view of accommodation zone, showing possible modes of deformation along the zone [from Faulds et al., 1990]. In cross sections and map views abbreviations are AZ, accommodation zone; upw, upper plate, west tilted domain; upe, upper plate, east tilted domain; lpw, lower plate, west tilted domain; lpe, lower plate, east tilted domain; up, untilted upper plate; lp, lower plate. (a) Oblique view of an accommodation zone. (b) Lower plate pulled out from beneath relatively stationary upper plate. Strike-slip offset along the zone occurs where lower plate of one domain is juxtaposed against either the upper or lower plate of adjacent domain. (c) Both upper and lower plates are translated significant lateral distances in opposite directions. Strike-slip offset along the zone occurs where (1) upper plates of opposing tilt-block domains are juxtaposed or (2) lower plates of adjacent domains are juxtaposed. The sense of strike-slip motion between the upper plates of opposing domains is opposite that between lower plates. (d) In cases where detachment geometry does not change across a zone (e.g., bulk pure-shear extension of continental lithosphere), strike-slip displacement is negligible along the zone unless the zone facilitates a change in the position of axes of extension.

slip displacement. In addition, the amount of Neogene tilting and/or flexing (i.e., rotation about a horizontal axis) of several large crystalline terranes near the accommodation zone is evaluated with paleomagnetic data from Late Cretaceous and Miocene intrusions. Data from the crystalline terranes are essential for assessing the lower or upper plate affinity of the terranes and the original dip of major low-angle normal faults. Knowledge of the magnitude and uniformity of tilting of large crystalline terranes may also facilitate accurate reconstructions of the extended region, mineral exploration efforts, investigations of magmatic processes, studies of Mesozoic/Proterozoic deformation and metamorphism, and ultimately assessment of models of continental extension. Furthermore, the process of crustal tilting bears on the interpretation of paleomagnetic data from crystalline terranes along continental margins, as tilting can affect paleomagnetic directions similar to that of significant latitudinal displacement and vertical axis rotation [Beck, 1980; Butler et al., 1989; May et al., 1989; Irving and Archibald, 1990].

Where flanked by major low-angle normal faults, large crystalline terranes have commonly been ascribed to a lower plate setting [Wernicke, 1985; Reynolds and Spencer, 1985; Howard and John, 1987; Davis and Lister, 1988]. Many of the bounding low-angle normal faults are presumed to have both nucleated and accommodated significant movement at gentle dips [e.g., Wernicke, 1985; Davis and Lister, 1988]. The large lateral extent of many terranes in a direction parallel to displacement appears to preclude appreciable wholesale tilting [Reynolds and Spencer, 1985; Davis and Lister, 1988]. However, isostatic rebound of tectonically denuded crystalline rocks [Spencer, 1984; Buck, 1988; Wernicke and Axen, 1988; Reynolds and Lister, 1990; Bartley et al., 1990] and/or multiple generations of low-angle normal faults (e.g., excisement/incisement processes of Davis and Lister [1988]) may produce appreciable flexing or tilting of portions of a lower plate (Figure 2). Thus it cannot be assumed that lower plate rocks have undergone negligible tilting or flexing. Nevertheless, the magnitude of flexing or tilting of much of the lower plate, especially its deeper structural levels, is probably much less than that of upper plate fault blocks in

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TABLE 1. Analytical Data, K/Ar Isotopic Age Determinations

| Sample Number | Location | Lithology | Tilt | Mineral | % Potassium Data | % Potassium Mean | Radiogenic Ar, pm/g Data | Radiogenic Ar, pm/g Mean | % Atm. Ar Data | % Atm. Ar Mean | Reported Date +/- Error, Ma |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 88-177 | SEM-WTD Opal Mountain | basalt flow | 15° W | feldspar groundmass | 0.916 | 0.934 | 21.39 | 21.33 | 21.8 | 22.2 | 13.13 +/- 0.27 |
| | | | | | 0.929 | | 21.28 | | 22.1 | | |
| | | | | | 0.938 | | 21.37 | | 22.2 | | |
| | | | | | 0.952 | | 21.27 | | 22.5 | | |
| 88-065 | CBM-WTD | basalt flow | 25° W | feldspar groundmass | 1.172 | 1.136 | 28.25 | 28.30 | 27.5 | 27.7 | 14.32 +/- 0.36 |
| | | | | | 1.121 | | 28.19 | | 28.1 | | |
| | | | | | 1.107 | | 28.34 | | 27.4 | | |
| | | | | | 1.143 | | 28.43 | | 27.8 | | |
| 88-063 | CBM-WTD | basalt flow | 44° W | feldspar groundmass | 1.211 | 1.218 | 30.06 | 30.08 | 16.1 | 15.7 | 14.10 +/- 0.37 |
| | | | | | 1.253 | | 30.38 | | 15.1 | | |
| | | | | | 1.201 | | 30.10 | | 15.8 | | |
| | | | | | 1.207 | | 28.78 | | 15.7 | | |
| 88-066 | CBM-WTD | ash flow tuff | 85° W | biotite | 7.286 | 7.277 | 207.1 | 208.2 | 7.9 | 7.7 | 16.43 +/- 0.36 |
| | | | | | 7.274 | | 208.3 | | 7.7 | | |
| | | | | | 7.272 | | 208.0 | | 7.8 | | |
| | | | | | | | 209.3 | | 7.4 | | |
| 88-068 | CBM-WTD Mount Perkins | rhyolite dike | -- | K-feldspar | 4.834 | 4.878 | 123.2 | 123.1 | 22.3 | 22.6 | 14.50 +/- 0.30 |
| | | | | | 4.908 | | 123.6 | | 22.3 | | |
| | | | | | 4.971 | | 122.6 | | 22.9 | | |
| | | | | | 4.798 | | 123.0 | | 22.8 | | |
| 88-181 | CBM-WTD Mount Perkins pluton | granodiorite | -- | biotite | 5.981 | 6.131 | 177.5 | 179.2 | 31.6 | 31.9 | 16.79 +/- 0.36 |
| | | | | | 6.185 | | 180.8 | | 31.2 | | |
| | | | | | 6.129 | | 179.6 | | 31.9 | | |
| | | | | | 6.226 | | 179.0 | | 32.8 | | |
| 88-176 | CBM-ETD Eldorado Canyon | basalt flow | 15° E | feldspar groundmass | 1.606 | 1.606 | 38.49 | 38.31 | 24.1 | 24.2 | 13.71 +/- 0.29 |
| | | | | | 1.612 | | 38.31 | | 24.1 | | |
| | | | | | 1.601 | | 38.25 | | 24.3 | | |
| | | | | | | | 38.20 | | 24.5 | | |
| 88-067 | CBM-ETD Eldorado Canyon | ash flow tuff | 45° E | biotite | 6.941 | 6.901 | 190.3 | 191.3 | 41.6 | 42.2 | 15.92 +/- 0.36 |
| | | | | | 6.887 | | 192.3 | | 41.4 | | |
| | | | | | 6.876 | | 194.2 | | 41.8 | | |
| | | | | | | | 188.5 | | 44.1 | | |
| 88-179 | SEM-ETD Aztec Wash/Knob Hill pluton | granodiorite | -- | biotite | 6.611 | 6.672 | 186.1 | 188.3 | 10.9 | 11.7 | 16.21 +/- 0.45 |
| | | | | | 6.763 | | 193.4 | | 11.6 | | |
| | | | | | 6.685 | | 184.6 | | 12.3 | | |
| | | | | | 6.629 | | 193.0 | | 11.6 | | |
| 88-180 | SEM-ETD Nelson pluton | granodiorite | -- | biotite | 7.001 | 7.082 | 208.7 | 208.8 | 53.6 | 53.8 | 16.93 +/- 0.48 |
| | | | | | 7.193 | | 209.5 | | 53.5 | | |
| | | | | | 7.108 | | 209.3 | | 53.8 | | |
| | | | | | 7.027 | | 207.8 | | 54.2 | | |
| 88-175 | CBM-ETD Radio Tower pluton | quartz monzonite | -- | muscovite | 8.634 | 8.656 | 1126 | 1122 | 14.2 | 14.7 | 73.3 +/- 1.5 |
| | | | | | 8.631 | | 1126 | | 14.8 | | |
| | | | | | 8.704 | | 1117 | | 14.8 | | |
| | | | | | | | 1119 | | 14.9 | | |

Sample number from Laboratory of Isotope Geochemistry, University of Arizona, Tucson. CBM, central Black Mountains; SEM, southern Eldorado Mountains; ETD, east-tilted domain; WTD, west-tilted domain. Potassium was analyzed on a Perkin-Elmer atomic absorption spectrophotometer (model 403) on at least three aliquots. Samples for Ar analyses were fused in induction-heated molybdenum crucibles suspended in 90 mm air-cooled pyrex fusion envelopes prebaked for 10-12 hours. Once thoroughly purified, the gas resulting from each fusion was divided into two or more aliquots for separate analyses by static mode using a 15.24 cm, 60 sector field, Nier-type gas source mass spectrometer. Dates were calculated using $^{40}K$ abundance and decay constants of Steiger and Jager [1977].

the highly extended, more proximal part of the detachment system.

On the other hand, steeply dipping nonconformities suggest that some large crystalline terranes correspond to lower structural levels of highly tilted, upper plate fault blocks rather than to the lower plate of a detachment terrane (Figure 2). Major low-angle normal faults that bound these terranes formed at moderate to steep dips and were subsequently rotated about a horizontal axis to gentle dips by dominolike block tilting [e.g., Morton and Black, 1975; Proffett, 1977; Chamberlin, 1978, 1983; Wernicke and Burchfiel, 1982; Miller et al., 1983; Davis, 1983, 1987]. The magnitude of tilting throughout such terranes generally mimics that of nearby fault blocks in the imbricate array.

Discriminating between a lower or upper plate setting is, however, difficult for some crystalline terranes. For example, the areal extent of some

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terranes does not rule out significant tilting. In addition, erosion or burial of layered supracrustal strata may obscure the origin of crystalline parts of highly tilted fault blocks. Where conventional field and petrologic studies fail, paleomagnetic methods may afford a useful tool with which to evaluate the amount of tilting or flexing [e.g., Beck, 1980; Faulds et al., 1988b; Dagley and Piper, 1989; Marquis and Irving, 1990].

Accurate determination of the magnitude of tilting (or flexing) of crystalline rocks with paleomagnetic data is contingent upon magnetic mineralogy and structural level of exposure. The survival of a well-defined pretilt characteristic remanent magnetization (ChRM) generally requires an assemblage of dominantly fine-grained (i.e., single and/or pseudo-single domain) low-Ti titanomagnetite of relatively high coercivity and moderate to high unblocking temperature, or hematite, the paragenesis of which can be determined. The presence of such magnetic grain assemblages does not guarantee, however, that a pretilt ChRM is preserved.

The preservation of a pretilt ChRM is less likely at deeper crustal levels [Beck, 1980], as the probability of magnetization unblocking increases at elevated temperatures. A magnetization may even be unblocked below short-term unblocking temperatures if elevated temperatures are maintained for long periods of time [Pullaiah et al., 1975]. Consequently, the ChRM of crystalline rocks may commonly be a viscous partial thermoremanent magnetization (VPTRM), acquired during uplift-induced cooling, rather than a primary thermoremanent magnetization (TRM). With other factors constant, the likelihood of VPTRM acquisition increases at deeper structural levels. A tilted fault block or flexed lower plate may record a complicated magnetization history with pretilt ChRMs at high structural levels, syntilt ChRMs at intermediate to deep structural levels, and even posttilt ChRMs at deep levels if tilting preceded significant uplift and erosion (Figure 3).

## GEOLOGIC SETTING

The northern Colorado River extensional corridor [Howard and John, 1987; Faulds et al., 1990] is a 50- to 100-km-wide region of moderately to severely extended crust situated between the relatively unextended Colorado Plateau to the east and Spring Range to the west (Figure 4). A major accommodation zone bisects the corridor, separating more than 15,000 km$^2$ of east tilted fault blocks to the north from 25,000 km$^2$ of dominantly west tilted blocks to the south [Faulds et al., 1988a, 1990]. This region occupies a critical position in the western Cordillera, as the belt of metamorphic core complexes in the southern Basin and Range province [Crittenden et al., 1980; Frost and Martin, 1982] terminates northward in its vicinity. The nature and location of this termination are poorly understood.

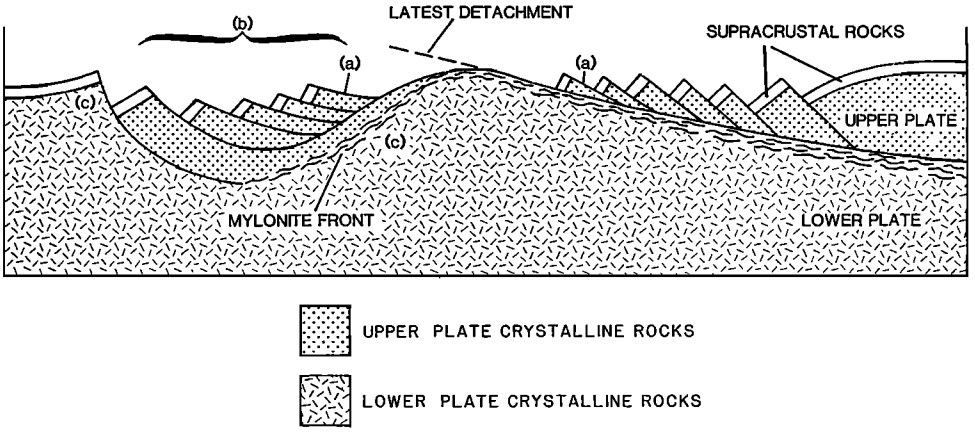



Fig. 2. Schematic cross section of a detachment terrane, after Davis and Lister [1988] and Reynolds and Lister [1990]; also see Wernicke and Axen [1988] and Buck [1988]. In highly extended regions, significant horizontal axis rotation of crystalline rocks can occur in several structural settings, including (a) lower structural levels of tilted upper plate fault blocks, (b) excised upper plate (i.e., part of an early upper plate that is transferred into the lower plate of a subsequent detachment), and (c) "primary" lower plate, where the isostatic response to tectonic denudation has produced significant flexing. The amount of flexing or tilting of most of the lower plate will probably be less than tilting of upper plate fault blocks in the proximal part of the detachment terrane.

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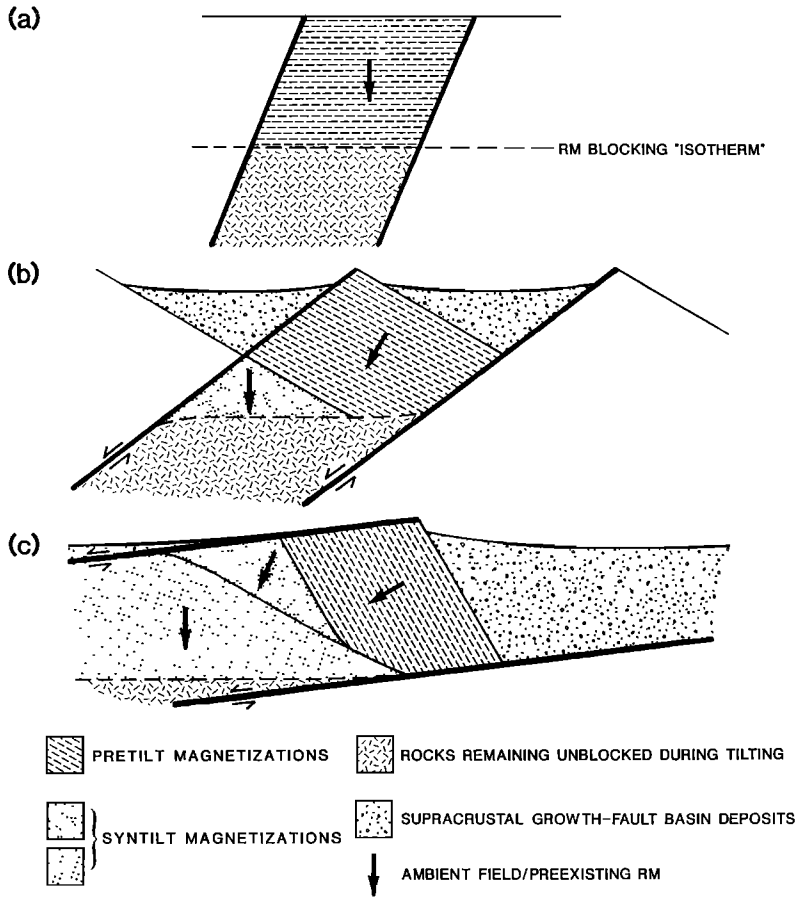



Fig. 3. Schematic history of magnetization blocking in (a) untilted, (b) moderately tilted, and (c) steeply tilted fault blocks. Different structural levels of tilted fault blocks may record distinct magnetizations as a result of magnetization blocking during tilting. The likelihood of preserving pretilt magnetizations is greater in the structurally higher level of a tilted block. Magnetizations in the deeper structural levels of highly tilted fault blocks may indicate little, if any, tilting. Voluminous magmatism roughly coeval with tilting would greatly affect the relative timing of magnetization blocking and patterns of viscous partial thermoremanant magnetization acquisition.

In the northern Colorado River extensional corridor, Tertiary volcanic and sedimentary strata generally rest directly on Proterozoic crystalline rocks. The distended Tertiary strata range from early to late Miocene in age. The Proterozoic basement is significantly intruded by Late Cretaceous to middle Miocene plutons and dike swarms.

Middle to late Miocene extension fragmented the extensional corridor into a complex array of tilted fault blocks [Anderson, 1971; Anderson et al., 1972; Weber and Smith, 1987; Faulds et al., 1988a, 1990]. Fault-block tilting, as indicated by layering in the Tertiary volcanic and sedimentary rocks, commonly exceeds 60°. The strike of both major normal faults and layering in tilted fault blocks suggests an east/west to east-northeast/west-southwest extension direction. Most of the extension near the accommodation zone occurred between 16.4 and 13.1 Ma, as evidenced by K/Ar isotopic age determinations on variably tilted volcanic strata in major growth-fault basins (Table 1). However, much of the tilting in the western Lake Mead area is bracketed between 13 and 9 Ma [Duebendorfer and Wallin, 1991].

The east-west trending accommodation zone (Figure 4) is a 10-km-wide belt of intermeshing conjugate normal faults that approximately parallels the extension direction. A west dipping detachment system beneath the east tilted domain [Wernicke, 1985; Weber and Smith, 1987; Duebendorfer et al., 1990] and east dipping detachment system in the west tilted domain [Davis et al., 1980; Spencer, 1985; Howard and John, 1987; Davis and Lister, 1988; Spencer and

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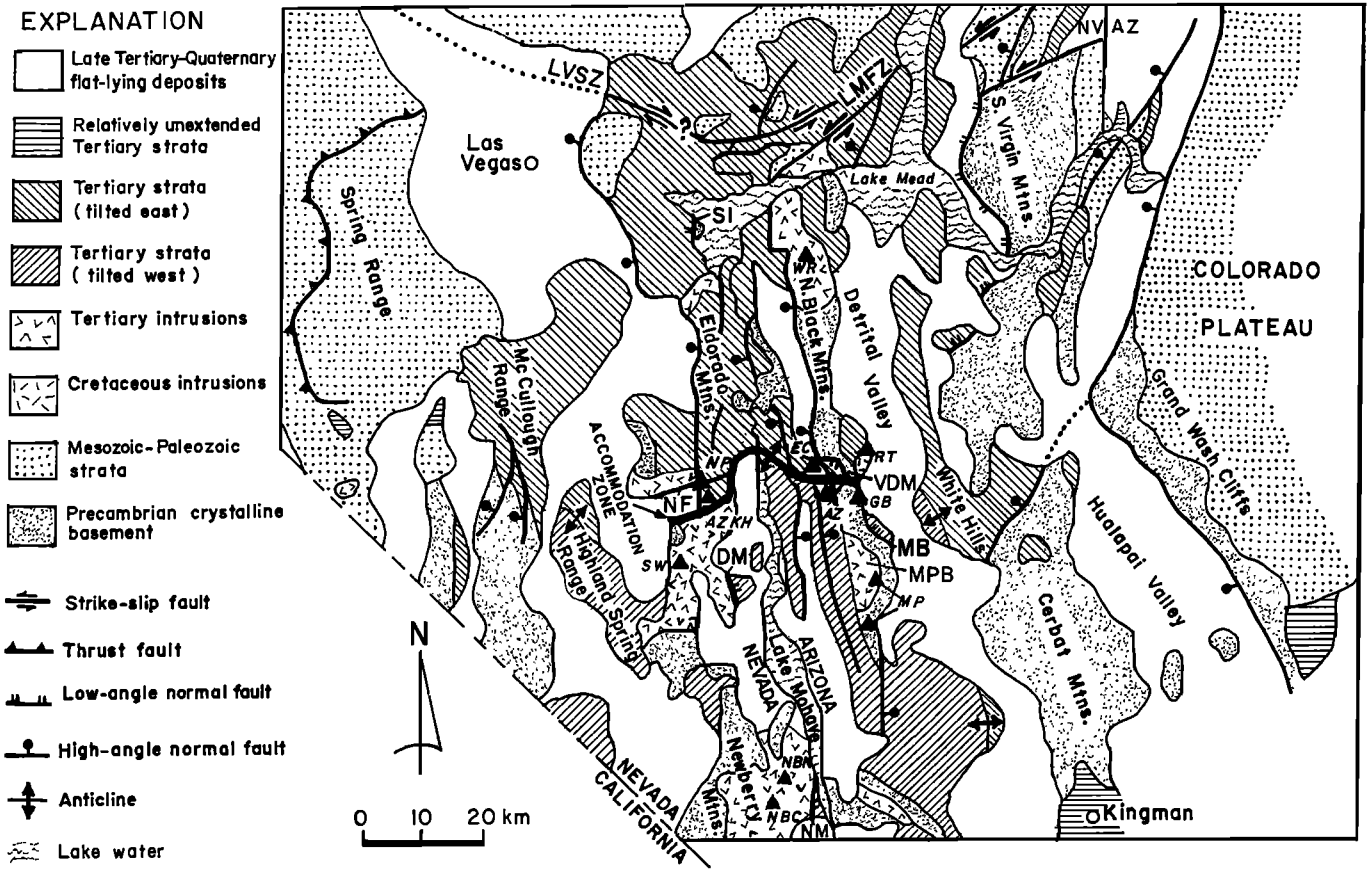



Fig. 4. Generalized geologic map of the northern Colorado River extensional corridor. DM, Dupont Mountain fault; LMFZ, Lake Mead fault zone; LVSZ, Las Vegas Valley shear zone; MB, Mockingbird Mine fault; MPB, Mount Perkins block; NF, Nelson fault; NM, Newberry Mountains detachment; SI, Saddle Island fault; VDM, Van Deemen Mine fault. The approximate locations of paleomagnetic sampling localities are shown by triangles: AZKH, Aztec Wash/Knob Hill pluton; AZ, accommodation zone; EC, Eldorado Canyon; GB, Gold Bug Mine area; MP, Mount Perkins area; NBC, central Newberry Mountains; NBN, northern Newberry Mountains; NP, Nelson pluton; RT, Radio Tower pluton; SW, southwestern part of southern Eldorado Mountains; WR, Wilson Ridge pluton.

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Reynolds, 1989] probably converge at the zone. Both the displacement on major normal faults and magnitude of block tilting progressively decrease toward the axial part of the zone. The tilt reversal is facilitated by transversely oriented, oblique-slip scissor faults on transverse segments (i.e., orthogonal to strike of tilted fault blocks) of the zone and minor folding on along-strike segments (i.e., parallel to strike of tilted blocks). The lack of consistent cross cutting relations between east and west dipping normal fault systems and K-Ar isotopic age determinations (Table 1) from variably tilted volcanic strata suggest penecontemporaneous east and west tilting near the zone. The deformational style in the zone is discussed in greater detail by Faulds et al. [1990].

Regional consistency in the average strike of both normal faults and layering in Tertiary strata suggests that little vertical axis rotation accompanied Miocene extension in most of the extensional corridor. However, in light of significant counterclockwise rotation in the southwestern Lake Mead area [Geissman, 1986], clockwise rotation along the Las Vegas Valley shear zone [Nelson and Jones, 1987; Sonder et al., 1989], variable vertical axis rotations in portions of the west tilted domain farther to the south [Wells and Hillhouse, 1989], and the possibility of distributed strike-slip faulting along the accommodation zone, the assumption of simple tilting was scrutinized using paleomagnetic data from Miocene volcanic strata (49 sites, 2 localities) in and near the accommodation zone (Figure 4). Paleomagnetic data were also collected from Miocene (?) dikes intruding upper plate crystalline rocks near the axis of the zone (7 sites, 1 locality) and Proterozoic gneiss and Miocene intrusions in a large west tilted fault block, the Mount Perkins block, directly south of the zone (12 sites, 2 localities) (Figure 4).

Large crystalline terranes in the northern Black, southern Eldorado, and Newberry mountains may correspond to lower plate rocks of the opposing detachment terranes or lower parts of tilted fault blocks. The northern Black Mountains terrane lies within the east tilted domain between the Lake Mead area to the north and accommodation zone to the south (Figure 4). A system of gently to steeply, west dipping normal faults bounds most of this terrane on the west. The southern Eldorado Mountains crystalline terrane extends southward from the southern part of the east tilted domain well into the west tilted domain, where it merges with the Newberry Mountains crystalline terrane (Figure 4). A major gently east dipping normal fault zone, the Dupont Mountain fault, bounds the southern Eldorado Mountains terrane on the east. The Dupont Mountain fault may represent the northward continuation of gently east dipping normal faults in the Newberry, Dead, and Sacramento mountains, described by Mathis [1982] and Spencer and Reynolds [1989] as major detachment faults. To assess the magnitude of horizontal axis rotation and lower and upper plate affinity, paleomagnetic data were collected from Late Cretaceous and Miocene intrusions in these crystalline terranes (47 sites, 7 localities).

## PALEOMAGNETIC SAMPLING AND METHODOLOGY

A portable drill was used to collect 5-10 core samples from each site. Cores were oriented with magnetic and sun compasses. Specimens were prepared for measurement using nonmagnetic saw blades. Measurements of remanent magnetization were carried out on a 2G Enterprises 760R superconducting rock magnetometer and a Schonstedt SSM-1A spinner magnetometer. Components of the remanent magnetization were isolated using progressive alternating-field (AF) or thermal demagnetization methods and principal component analysis [e.g., Kirschvink, 1980]. Site and locality mean directions were calculated using Fisher [1953] statistics. Results from sites with $\alpha_{95} > 25^{\circ}$ were omitted from further analysis.

In situ site mean directions from Miocene volcanic strata of the central Black Mountains were corrected for tilt by rotating about the local strike line. Orientations of basal flow contacts, eutaxitic foliation in ash flow tuffs, and/or bedding in intercalated sedimentary rocks were used to determine the structural attitude at each site. The fold test of Graham [1949] was applied to site mean directions from tilted volcanic strata. Scissorslike faulting in the accommodation zone suggests, however, that parts of the central Black Mountains were tilted about plunging axes. Simple tilt corrections in such areas can generate significant declination errors [MacDonald, 1980; Chan, 1988]. To average the effects of scissor faults, volcanic flows were sampled from several fault blocks. Vertical axis rotations were assessed by comparing tilt-corrected locality mean directions with expected Miocene geomagnetic field directions calculated from the paleomagnetic poles of Irving and Irving [1982] and Mankinen et al. [1987].

Tilt corrections were not applied to paleomagnetic data from crystalline rocks, because paleohorizontal could not be accurately determined. The magnitude of tilting of the crystalline rocks was inferred by assuming that discordant site mean directions resulted entirely from tilting and/or flexing about horizontal axes.

Well-defined magnetizations of probable geologic significance, as evidenced by high laboratory unblocking temperatures ($T_{ub}$) and coercivity spectra, were isolated in some crystalline rocks. In many rocks,

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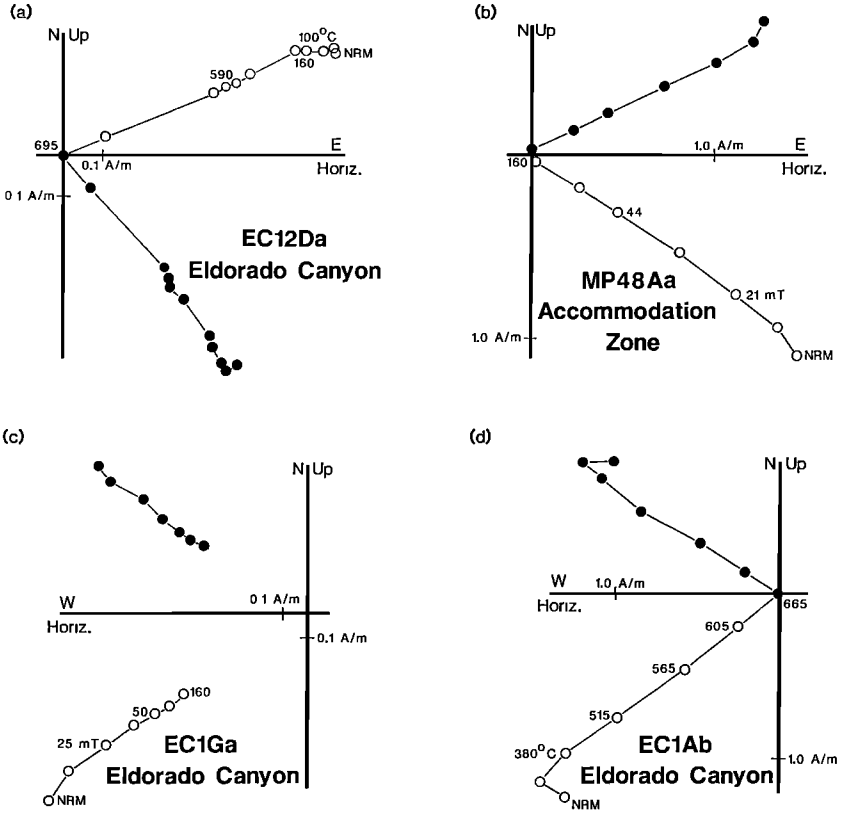



Fig. 5. Representative orthogonal demagnetization diagrams from Miocene volcanic rocks showing the progressive demagnetization behavior of individual specimens during alternating field (AF) and thermal demagnetization. The endpoint of the magnetization vector, over a series of demagnetization steps, is projected simultaneously onto both the horizontal (solid symbols) and vertical (open symbols) planes. In situ (geographic) coordinates are always used. Peak demagnetizing inductions are given in milliTesla (1 mT = 10 Oe) and temperatures in degrees Celsius. In each case, the direction of the magnetization component, characteristic of an individual site or locality, is isolated over several demagnetization steps. The associated maximum angular deviation (MAD) is also calculated. (a) Andesite flow, Eldorado Canyon, central Black Mountains, site EC12; direction of ChRM: 140/-23 (declination, inclination), MAD = 1.2°. (b) AF demagnetization, basalt flow, accommodation zone, west tilted domain, central Black Mountains, site MP48; ChRM: 63/35, MAD = 1.3°. (c) AF demagnetization, basaltic andesite flow, Eldorado Canyon, central Black Mountains, site EC1; ChRM: 303/34, MAD = 1.0°. (d) Thermal demagnetization, basaltic andesite flow, Eldorado Canyon, central Black Mountains, site EC1; ChRM: 303/39, MAD = 1.6°.

however, low median destructive inductions and/or $T_{ub}$ values, as well as erratic behavior in progressive demagnetization, indicate the presence of abundant, coarse-grained multidomain titanomagnetite. The similarity in the direction of the normal-polarity Miocene and present-day fields precludes straightforward discrimination between magnetizations of uplift and recent viscous origin. In an attempt to resolve this problem, we sampled individual intrusions in many areas and, where possible, collected several rock types at a single locality. Magnetizations of northerly declination and moderate positive inclination, completely removed at low peak inductions (< about 30 mT) and low $T_{ub}$ values (< approximately 300°C), were interpreted as recently acquired viscous magnetizations. Most sites collected in Proterozoic gneiss and 60% of the sites from Miocene intrusions were dominated by a magnetization of probable viscous origin.

## PALEOMAGNETIC RESULTS AND INTERPRETATIONS

### Central Black Mountains: Volcanic Rocks

A total of 54 sites were collected from Miocene volcanic strata in the

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TABLE 2. Paleomagnetic Data, Volcanic Rocks, Accommodation Zone

| | | | | In Situ | | | | Corrected | |
|---|---|---|---|---|---|---|---|---|---|
| Locality | Site | Lithology | $N/N_0$ | D | I | $a_{95}$ | k | D | I |
| ETD | MP28 | basaltic andesite | 8/8 | 287 | 24 | 5.8 | 92 | 310 | 70 |
| ETD | MP30 | basaltic andesite | 10/10 | 297 | 17 | 3.8 | 161 | 316 | 47 |
| ETD | MP31 | basaltic andesite | 9/10 | 295 | 13 | 10.2 | 26 | 311 | 39 |
| WTD | MP32 | basaltic andesite | 7/8 | 28 | 56 | 2.5 | 449 | 333 | 79 |
| WTD | MP33 | basaltic andesite | 9/9 | 15 | 65 | 4.5 | 131 | 335 | 82 |
| WTD | MP34 | basaltic andesite | 8/8 | 359 | 33 | 3.9 | 164 | 333 | 62 |
| WTD | MP35 | basaltic andesite | 9/9 | 30 | 45 | 7.1 | 56 | 35 | 67 |
| WTD | MP37 | basalt | 7/8 | 322 | 55 | 4.5 | 178 | 318 | 50 |
| WTD | MP38 | basalt | 8/8 | 43 | 58 | 2.5 | 496 | 14 | 64 |
| WTD | MP39 | basaltic andesite | 7/8 | 348 | 18 | 6.1 | 102 | 338 | 41 |
| ETD | MP40 | basaltic andesite | 4/8 | 13 | 58 | 4.1 | 524 | 24 | 51 |
| WTD | MP41 | basaltic andesite | 6/8 | 343 | 46 | 4.6 | 216 | 339 | 51 |
| WTD | MP42 | basaltic andesite | 7/8 | 339 | 52 | 10.9 | 32 | 335 | 56 |
| ETD | MP43 | basaltic andesite | 7/8 | 352 | 50 | 4.1 | 219 | 18 | 52 |
| ETD | MP44 | basaltic andesite | 8/9 | 355 | 44 | 4.2 | 177 | 359 | 41 |
| WTD | MP47 | basalt | 8/8 | 64 | 44 | 6.3 | 78 | 14 | 56 |
| WTD | MP48 | basalt | 8/8 | 62 | 37 | 6.6 | 71 | 23 | 60 |
| WTD | MP49 | basalt | 7/8 | 59 | 61 | 3.2 | 358 | 345 | 49 |
| WTD | MP50 | basalt | 8/8 | 63 | 62 | 4.8 | 133 | 334 | 76 |
| WTD | MP51 | basalt | 8/8 | 54 | 41 | 6.6 | 72 | 7 | 63 |
| WTD | MP52 | ash flow tuff | 9/9 | 25 | -12 | 5.7 | 83 | 16 | 24 |
| WTD | MP56 | basalt | 7/8 | 11 | 55 | 4.8 | 160 | 4 | 64 |
| WTD | MP57 | basalt | 7/8 | 63 | 44 | 4.4 | 190 | 42 | 63 |
| WTD | MP58 | basalt | 9/9 | 66 | 50 | 6.4 | 66 | 29 | 75 |
| WTD | MP59 | basalt | 9/9 | 25 | 46 | 6.7 | 61 | 3 | 53 |
| ETD | MP61 | basaltic andesite | 8/8 | 257 | 44 | 9.2 | 38 | 60 | 75 |
| ETD | MP62 | basaltic andesite | 6/6 | 121 | 3 | 3.6 | 343 | 140 | -56 |
| ETD | MP63 | basaltic andesite | 7/8 | 297 | 25 | 3.8 | 249 | 347 | 64 |

$N/N_0$ is number of samples calculated for site mean/number of samples measured; D, Declination; I, Inclination; $a_{95}$, semi-angle of cone at 95% confidence level; k, estimate of precision [Fisher, 1953]; ETD, east-tilted domain; WTD, west-tilted domain.

central Black Mountains at two localities. The sites were obtained from either the accommodation zone or Eldorado Canyon, approximately 2.5 km north of the zone (Figure 4). Sampled rock types include both pretilting early Miocene and syntilting middle Miocene basaltic andesite flows, ash flow tuffs, and basalt flows.

Demagnetization behavior of the Miocene volcanic rocks generally varies with position in the stratigraphic section. ChRMs in basaltic andesite flows in lower to middle parts of the section reside largely in hematite, as demonstrated by median destructive inductions > 160 mT and $T_{ub}$ values up to 680°C (Figure 5a). The hematite probably formed by deuteric or hydrothermal alteration prior to the onset of major tilting at approximately 16.4 Ma. In contrast, median destructive inductions of 30-160 mT and $T_{ub}$ values of 450°-665°C indicate that ChRMs in basalt and basaltic andesite flows in the upper half of the section reside in both fine-grained titanomagnetite and hematite (Figures 5b, 5c, and 5d).

Magnetization acquisition in the Miocene volcanic section predated tilting. Tilt corrections for 28 sites in the accommodation zone locality reduce $a_{95}$ from 14.5° to 6.4° and increase k from 4.5 to 17.4 (Tables 2 and 4, and Figures 6a and 6b). The fold test of Graham [1949] is significant at the 99% confidence level [McElhinny, 1964]. Tilt corrections result in little change of angular dispersion for 21 site mean directions from Eldorado Canyon because

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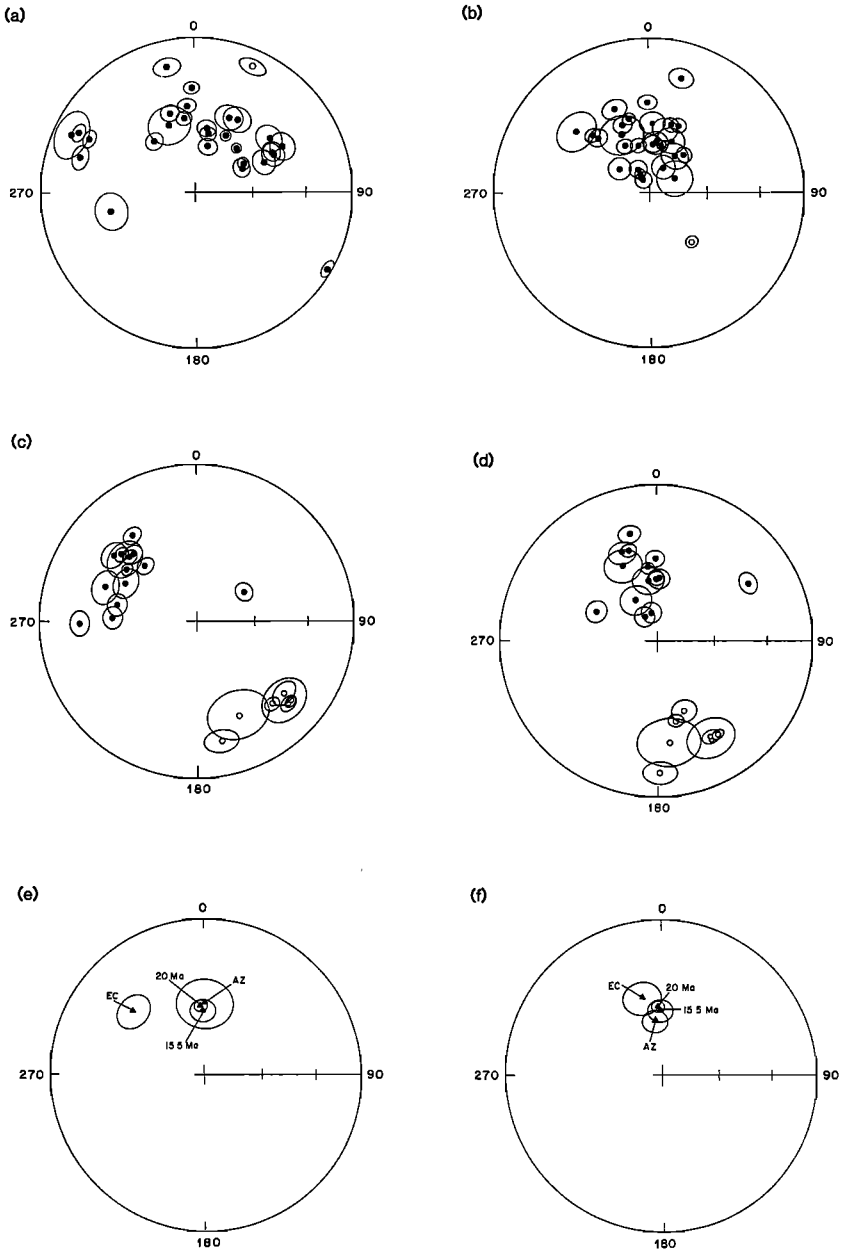



Fig. 6. Equal-area projections of site mean magnetizations, Miocene volcanic rocks, central Black Mountains. Solid symbols are projections on lower hemisphere; open symbols, upper hemisphere. The projected cones of 95% confidence are also shown for each site. (a) In situ site mean magnetizations, accommodation zone locality, $N/N_0$ = 28/32 sites. (b) Tilt-corrected site mean magnetizations, accommodation zone locality, $N/N_0$ = 28/32 sites. (c) In situ site mean magnetizations, Eldorado Canyon locality, $N/N_0$ = 21/23 sites. (d) Tilt-corrected site mean magnetizations, Eldorado Canyon locality, $N/N_0$ = 21/23 sites. (e) In situ locality means, AZ, accommodation zone locality; EC, Eldorado Canyon locality; 15.5 Ma, expected middle Miocene direction derived from the paleomagnetic pole of Mankinen et al. [1987]; 20 Ma, expected Miocene direction derived from the paleomagnetic pole of Irving and Irving [1982]. (f) Tilt-corrected locality means, AZ, accommodation zone locality; EC, Eldorado Canyon locality; 15.5 Ma, expected middle Miocene direction from paleomagnetic pole of Mankinen et al. [1987]; 20 Ma, expected Miocene direction from paleomagnetic pole of Irving and Irving [1982].

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little variation in structural attitude occurs at this locality (Tables 3 and 4 and Figures 6c and 6d).

The tilt-corrected, locality mean directions from the accommodation zone and Eldorado Canyon overlap expected Miocene directions [Mankinen et al., 1987; Irving and Irving, 1982] at the 95% confidence level (Figures 6e and 6f). Slight discordances between the two locality mean directions and expected directions (Table 4) may reflect incomplete averaging of geomagnetic secular variation. Virtual geomagnetic pole (VGP) angular standard deviations for the accommodation zone and Eldorado Canyon localities are 25.6° and 24.0°, respectively. These values exceed those

TABLE 3. Paleomagnetic Data, Volcanic Rocks, Eldorado Canyon

| | | | | In Situ | | | | Corrected | |
|---|---|---|---|---|---|---|---|---|---|
| Locality | Site | Lithology | $N/N_0$ | D | I | $a_{95}$ | k | D | I |
| ETD | EC1 | basaltic andesite | 7/7 | 299 | 47 | 7.6 | 63 | 352 | 58 |
| ETD | EC2 | basaltic andesite | 9/9 | 308 | 43 | 2.8 | 336 | 353 | 51 |
| ETD | EC3 | basaltic andesite | 8/8 | 317 | 40 | 5.1 | 126 | 2 | 57 |
| ETD | EC4 | basaltic andesite | 8/8 | 315 | 40 | 3.7 | 222 | 0 | 57 |
| ETD | EC5 | basaltic andesite | 7/7 | 292 | 37 | 8.1 | 58 | 333 | 66 |
| ETD | EC6 | basaltic andesite | 6/7 | 283 | 47 | 5.5 | 148 | 348 | 75 |
| ETD | EC7 | basaltic andesite | 8/8 | 274 | 45 | 5.1 | 119 | 332 | 76 |
| ETD | EC8 | basaltic andesite | 8/8 | 310 | 32 | 6.8 | 67 | 339 | 39 |
| ETD | EC9 | basaltic andesite | 8/8 | 313 | 36 | 3.9 | 205 | 343 | 39 |
| ETD | EC10 | basaltic andesite | 8/8 | 324 | 32 | 4.7 | 140 | 346 | 30 |
| ETD | EC11 | dacite | 8/8 | 129 | -29 | 6.2 | 82 | 159 | -50 |
| ETD | EC12 | andesite | 5/5 | 138 | -29 | 3.9 | 388 | 167 | -46 |
| ETD | EC13 | basaltic andesite | 7/8 | 58 | 60 | 5.1 | 147 | 58 | 31 |
| ETD | EC14 | basaltic andesite | 8/8 | 270 | 26 | 5.7 | 94 | 296 | 55 |
| ETD | EC15 | basalt | 4/5 | 312 | 39 | 9.5 | 95 | 336 | 46 |
| ETD | EC16 | ash flow tuff | 5/5 | 156 | -35 | 14.9 | 27 | 173 | -34 |
| ETD | EC17 | basaltic andesite | 7/7 | 318 | 49 | 4.5 | 179 | 359 | 46 |
| ETD | EC18 | basaltic andesite | 5/7 | 211 | 53 | 9.3 | 69 | --- | -- |
| ETD | EC19 | basaltic andesite | 7/7 | 199 | 53 | 12.5 | 24 | --- | -- |
| ETD | EC20 | basaltic andesite | 6/8 | 169 | -23 | 7.5 | 81 | 179 | -16 |
| ETD | EC22 | basaltic andesite | 7/8 | 132 | -26 | 11.5 | 28 | 151 | -29 |
| ETD | EC23 | basaltic andesite | 8/8 | 131 | -22 | 2.7 | 436 | 147 | -29 |
| ETD | EC24 | basaltic andesite | 7/7 | 132 | -23 | 4.2 | 208 | 151 | -30 |

Notation same as in Table 2.

TABLE 4. Group Site Means/Expected Directions, Volcanic Rocks, Central Black Mountains

| | $N/N_0$ | D | I | $a_{95}$ | k | VGP ASD | R | ΔR | F | ΔF |
|---|---|---|---|---|---|---|---|---|---|---|
| Accommodation Zone | | | | | | | | | | |
| In situ, group site mean | 28/32 | 0.7 | 51.6 | 14.5 | 4.5 | -- | -- | -- | -- | -- |
| Corrected, group site mean | 28/32 | 353.4 | 61.3 | 6.7 | 17.4 | 25.6 | -5.3 | 11.6 | -4.9 | 4.8 |
| Eldorado Canyon | | | | | | | | | | |
| In situ, group site mean | 21/23 | 311.6 | 38.7 | 8.9 | 13.6 | -- | -- | -- | -- | -- |
| Corrected, group site mean | 21/23 | 346.3 | 47.8 | 9.1 | 13.5 | 24.0 | -12.4 | 11.1 | 8.6 | 3.6 |
| Expected direction at 15.5 Ma [Mankinen et al., 1987] | -- | 358.7 | 56.4 | 6.3 | -- | -- | -- | -- | -- | -- |
| Expected direction at 20 Ma [Irving and Irving, 1982] | -- | 358.1 | 53.9 | 3.1 | -- | -- | -- | -- | -- | -- |

VGP ASD, virtual geomagnetic pole angular standard deviation; R, amount of rotation ((-) = counterclockwise); F, amount of flattening; ΔR and ΔF, 95% confidence limits for rotation and flattening [Demarest, 1983]. Other notation as in Table 2.

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expected for paleosecular variation in the late Cenozoic [Merrill and McElhinny, 1983]. However, only one reverse-polarity flow was sampled at the accommodation zone locality. Tilting about a plunging axis may also contribute to dispersion. Nonetheless, these data imply that the central Black Mountains have experienced little vertical axis rotation during Neogene time.

Because the Miocene structural grain throughout the northern Colorado River extensional corridor generally mimics that in the central Black Mountains, we suggest that vertical axis rotation during Miocene extension is also negligible in sampled areas of the northern Black, southern Eldorado, and Newberry mountains. Discordance between in situ paleomagnetic data from crystalline rocks in these regions and expected directions are therefore attributed to tilting about horizontal axes rather than rotations about vertical axes.

## Central Black Mountains: Crystalline Rocks

Mafic to intermediate composition Miocene (?) dikes cutting Proterozoic gneiss and amphibolite near the axis of the accommodation zone (Gold Bug locality) yielded a ChRM of northerly declination and positive inclination (Table 5 and Figure 7a). Median destructive inductions generally range from 15 to 45 mT (Figure 8a). The crystalline terrane hosting the dikes appears to be a small upper plate horst. The in situ paleomagnetic directions are in accord with minor (i.e., < 25°) tilting observed in Tertiary strata in the axial part of the accommodation zone.

Proterozoic gneiss, middle Miocene dikes, and the 16.8 +/- 0.4 Ma Mount Perkins pluton (Table 1, sample 88-181) were sampled in the west tilted Mount Perkins block (Figure 4). Moderate to high median destructive inductions and $T_{ub}$ indicate that the ChRM in the Proterozoic gneiss and Miocene intrusions reside in fine-grained titanomagnetite. Median destructive inductions and $T_{ub}$ in the Proterozoic gneiss range from 20 to 40 mT and 400°C to 550°C, respectively, as compared to 70 to 100 mT and 500°C to 550°C in the Mount Perkins pluton (Figures 8b, 8c, and 8d).

The Mount Perkins pluton, Proterozoic gneiss, and a 14.5 +/- 0.3 Ma (Table 1, 88-068) felsic dike swarm yielded a ChRM of southwesterly declination and moderate to shallow negative inclination (or antipodes) (Figure 7b). The anisotropy of magnetic susceptibility in the gneiss and pluton appears inadequate to deflect the ChRM [Faulds et al., 1987]. This assertion depends, however, on the domain state of the magnetic phases [Potter and Stephenson, 1988]. Proterozoic gneiss near the pluton was probably remagnetized during middle Miocene magmatism, as evidenced by a ChRM statistically indistinguishable from that of the pluton. We interpret the ChRM (shallow inclination) in Proterozoic gneiss more than 1 km from the pluton margin as a Late Cretaceous to middle Miocene VPTRM that either predates or is penecontemporaneous with the onset of major extension.

The paleomagnetic data indicate 40°-90° of west tilting of the Mount Perkins area about a north-northwest trending axis

TABLE 5. Paleomagnetic Data, In Situ, Crystalline Rocks

| Locality | Site | Lithology | $N/N_0$ | D | I | $a_{95}$ | k |
|---|---|---|---|---|---|---|---|
| | | Central Black Mountains | | | | | |
| Gold Bug Mine area | GB1 | intermediate dike | 5/7 | 6.1 | 58.8 | 8.4 | 84.2 |
| | GB2 | mafic dike | 7/7 | 344.4 | 49.1 | 14.1 | 19.3 |
| | GB3 | mafic dike | 5/6 | 3.8 | 64.2 | 11.1 | 48.4 |
| | GB4 | mafic dike | 6/8 | 13.9 | 56.7 | 15 3 | 20.2 |
| | GB6 | mafic dike | 5/6 | 349.7 | 52.5 | 12.9 | 36.1 |
| | GB7 | mafic dike | 4/4 | 2 8 | 58.2 | 9 6 | 92 4 |
| | GB8 | mafic dike | 7/7 | 1.7 | 58 6 | 6.7 | 83.1 |
| Mount Perkins block | BCR1 | felsic dike | 5/5 | 41.9 | 13.1 | 11.2 | 31.1 |
| | MP4 | felsic dike | 8/8 | 9.3 | 23 3 | 7.7 | 41.2 |
| | MP6 | mafic dike | 6/6 | 222.3 | -28.1 | 7.5 | 80.1 |
| | MP7 | granodiorite pluton | 7/7 | 216.5 | -32.7 | 2.6 | 548.1 |
| | MP9 | granodiorite pluton | 6/6 | 213.6 | -32.1 | 4.9 | 188.1 |
| | MP12 | granodiorite pluton | 6/8 | 220.1 | -24.9 | 11.3 | 36 1 |
| | MP17 | granodiorite pluton | 6/6 | 229 9 | -31 2 | 3.1 | 491.1 |
| | MP18 | granodiorite pluton | 6/7 | 227.1 | -28.4 | 7.8 | 53.1 |
| | BCR2 | Proterozoic gneiss | 7/8 | 224.5 | -4 8 | 10.1 | 27.6 |
| | BCR4 | Proterozoic gneiss | 8/8 | 26.1 | -18.6 | 8.1 | 38.1 |
| | MP22 | Proterozoic gneiss | 6/6 | 217.5 | -31.6 | 7.1 | 64.9 |
| | MP24 | Proterozoic gneiss | 7/8 | 235.8 | -4.6 | 8.1 | 42.4 |
| | | Northern Black Mountains Crystalline Terrane | | | | | |
| Radio Tower pluton (southern part of terrane) | RT1 | quartz monzonite | 6/7 | 270.4 | 32.8 | 7 1 | 91.1 |
| | RT2 | quartz monzonite | 7/7 | 266.1 | 34.5 | 1.9 | 992.9 |
| | RT4 | quartz monzonite | 6/7 | 277.8 | 31.1 | 4.9 | 188.7 |
| | RT6 | quartz monzonite | 5/5 | 264 3 | 15.2 | 8.2 | 88.7 |
| | RT15 | quartz monzonite | 8/10 | 263.2 | 37.7 | 3.8 | 214.2 |
| | RT16 | quartz monzonite | 6/7 | 263.6 | 37 4 | 4.1 | 266.4 |
| | RT17 | quartz monzonite | 6/7 | 275.4 | 32.7 | 6.7 | 100.6 |
| | RT18 | quartz monzonite | 7/8 | 271.9 | 30.9 | 6.6 | 85.2 |
| | RT19 | quartz monzonite | 6/9 | 269.2 | 42.4 | 6.1 | 120.4 |
| | RT20 | quartz monzonite | 7/8 | 276.3 | 37.1 | 5.2 | 131.2 |
| Wilson Ridge area (northern part of terrane) | WR1 | intermediate dike | 4/6 | 154.8 | -49.1 | 9.1 | 103 5 |
| | WR2 | intermediate dike | 7/7 | 188.6 | -61.3 | 4.6 | 170 1 |
| | WR3 | intermediate dike | 6/9 | 159.3 | -46.6 | 8.2 | 67.1 |
| | WR9 | granodiorite pluton | 7/8 | 191.9 | -51.8 | 10.1 | 37.2 |
| | WR12 | granodiorite pluton | 7/8 | 14.7 | 61 5 | 5.6 | 115.5 |
| | WR13 | granodiorite pluton | 8/9 | 350.6 | 57.1 | 4.5 | 149.9 |
| | WR15 | mafic dike | 6/7 | 122.3 | -40.5 | 5.7 | 140.8 |
| | WR17 | granodiorite pluton | 8/9 | 15.7 | 60.7 | 4.2 | 172.9 |
| | WR19 | granodiorite pluton | 6/8 | 22.1 | 45.9 | 9.1 | 56.9 |
| | WR20 | granodiorite pluton | 5/8 | 9.2 | 57.4 | 9 9 | 61.1 |
| | WR32 | diorite pluton | 5/9 | 193.9 | -47.2 | 11.5 | 45.4 |
| | WR34 | diorite pluton | 5/8 | 190.4 | -47.2 | 16.3 | 22.9 |
| | WR35 | diorite pluton | 5/8 | 15 3 | 56.1 | 2.8 | 773.9 |
| | | Southern Eldorado Mountains Crystalline Terrane | | | | | |
| Dike Swarm (southern part of terrane) | EM40 | intermediate dike | 7/7 | 213.9 | -25.5 | 5.6 | 117.8 |
| | EM42 | intermediate dike | 5/6 | 241.5 | -37.2 | 11.7 | 43.9 |
| | EM44 | intermediate dike | 5/7 | 234.5 | -8.3 | 7.5 | 104.8 |
| | EM46 | intermediate dike | 4/6 | 226.2 | -25 3 | 13.5 | 47 3 |
| | EM49 | intermediate dike | 7/9 | 212.6 | -22.9 | 8.4 | 52.4 |
| Nelson Pluton (northern part of terrane) | NP4 | granodiorite | 9/9 | 315.1 | 45.1 | 11.9 | 19.7 |
| | NP6 | granodiorite | 8/8 | 319.1 | 20.2 | 7.2 | 60.1 |
| | NP8 | granodiorite | 9/9 | 295.9 | 27.9 | 4.1 | 160.5 |
| | NP11 | granodiorite | 8/8 | 312.8 | 43.4 | 7 4 | 57.7 |
| | NP14 | granodiorite | 8/8 | 301.8 | 23.2 | 6.2 | 82.1 |
| | NP15 | granodiorite | 5/5 | 312.7 | 29.1 | 8.2 | 87.7 |
| | NP16 | granodiorite | 8/8 | 311.1 | 34.1 | 5.9 | 87.8 |
| Aztec Wash/Knob Hill Pluton (northern part of terrane) | EM23 | granodiorite | 8/9 | 101.2 | -42.1 | 5 1 | 123.1 |
| | EM33 | granodiorite | 8/8 | 100.5 | 8.7 | 9.9 | 32 4 |
| | EM34 | granodiorite | 7/7 | 111 7 | -3.5 | 11 1 | 31.3 |
| | | Newberry Mountains Crystalline Terrane | | | | | |
| Central part of terrane | NB3 | felsic dike | 7/8 | 7.8 | 52.1 | 10.3 | 29.8 |
| | NB4 | felsic dike | 8/9 | 355.7 | 52.4 | 4.1 | 157.6 |
| | NB5 | granodiorite pluton | 5/6 | 358.2 | 54.9 | 15.8 | 24.2 |
| | NB6 | mafic dike | 8/8 | 2.7 | 53.1 | 4.1 | 182.2 |
| | NB7 | intermediate dike | 10/11 | 6.6 | 53.2 | 5.5 | 76.5 |
| | NB8 | intermediate dike | 10/10 | 1.1 | 56.1 | 3.7 | 169.1 |
| Northern part of terrane | NB9 | Proterozoic granite | 6/7 | 32.1 | 40.9 | 7.6 | 79.2 |
| | NB10 | mafic dike | 9/9 | 37.2 | 40.7 | 8.8 | 35 4 |
| | NB11 | mafic dike | 12/12 | 39.5 | 30 1 | 5.9 | 54.1 |

Notation same as in Table 2.

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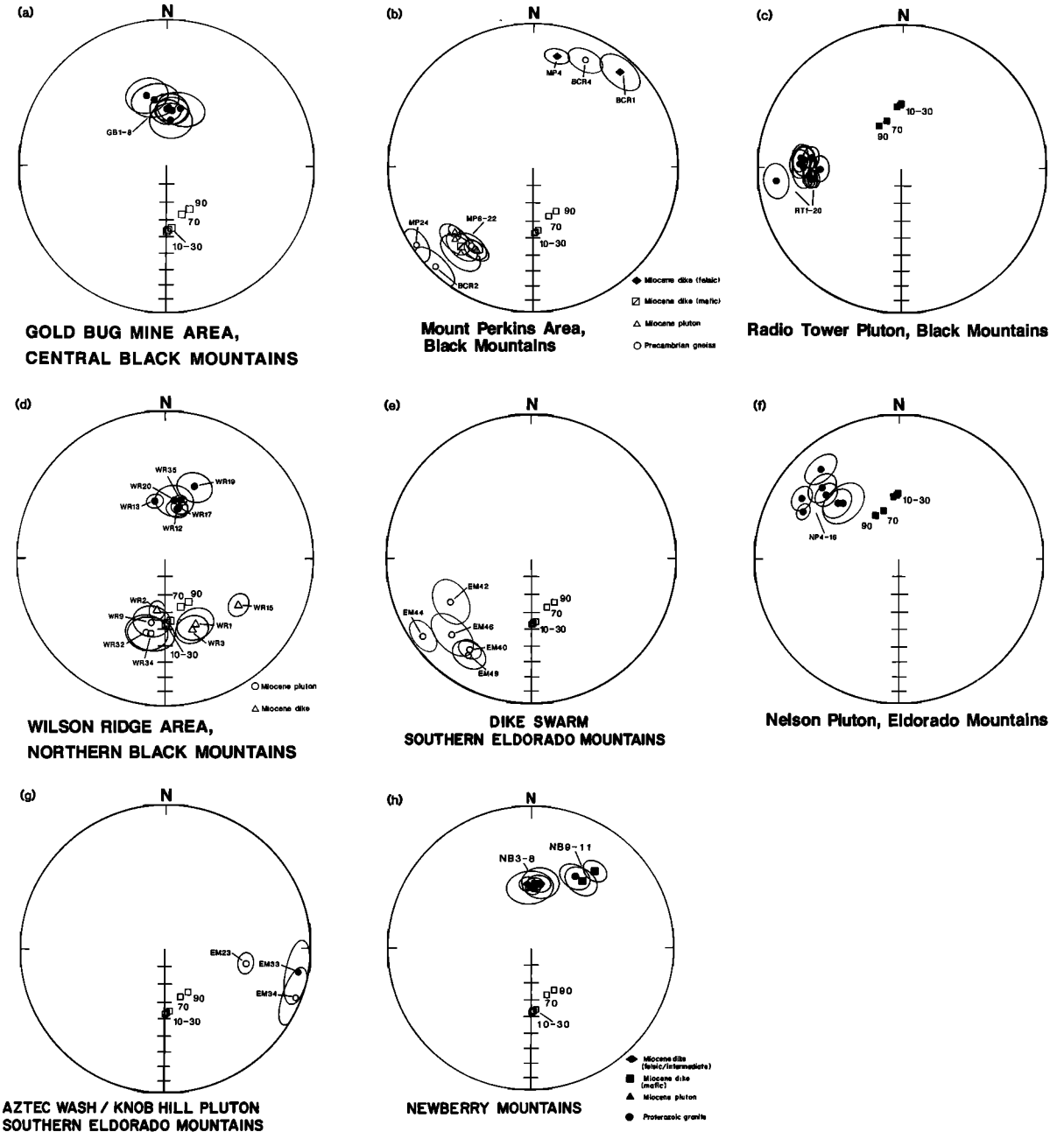



Fig. 7. Equal-area projections of in situ, site mean magnetizations of crystalline rocks. Solid symbols are projections on lower hemisphere; open symbols, upper hemisphere. The projected cones of 95% confidence are also shown for each site. Expected directions for 10, 20, 30, 70, and 90 Ma (based on poles of Irving and Irving [1982]) are shown as squares. (a) Miocene (?) dikes, Gold Bug Mine area, eastern part of central Black Mountains. (b) Mount Perkins block, central Black Mountains. (c) 73.3 +/- 1.5 Ma Radio Tower pluton, southern part of northern Black Mountains crystalline terrane. (d) 13.5 +/- 0.4 Ma Wilson Ridge pluton, northern part of northern Black Mountains crystalline terrane. (e) Miocene (?) dikes, southwestern part of southern Eldorado Mountains crystalline terrane. (f) 16.9 +/- 0.5 Ma Nelson pluton, central Eldorado Mountains. (g) 16.2 +/- 0.5 Ma Aztec Wash/Knob Hill pluton, northern part of southern Eldorado Mountains crystalline terrane. (h) Newberry Mountains crystalline terrane. Sites NB3-NB8 from central part of terrane; sites NB9-NB11 from northern part.

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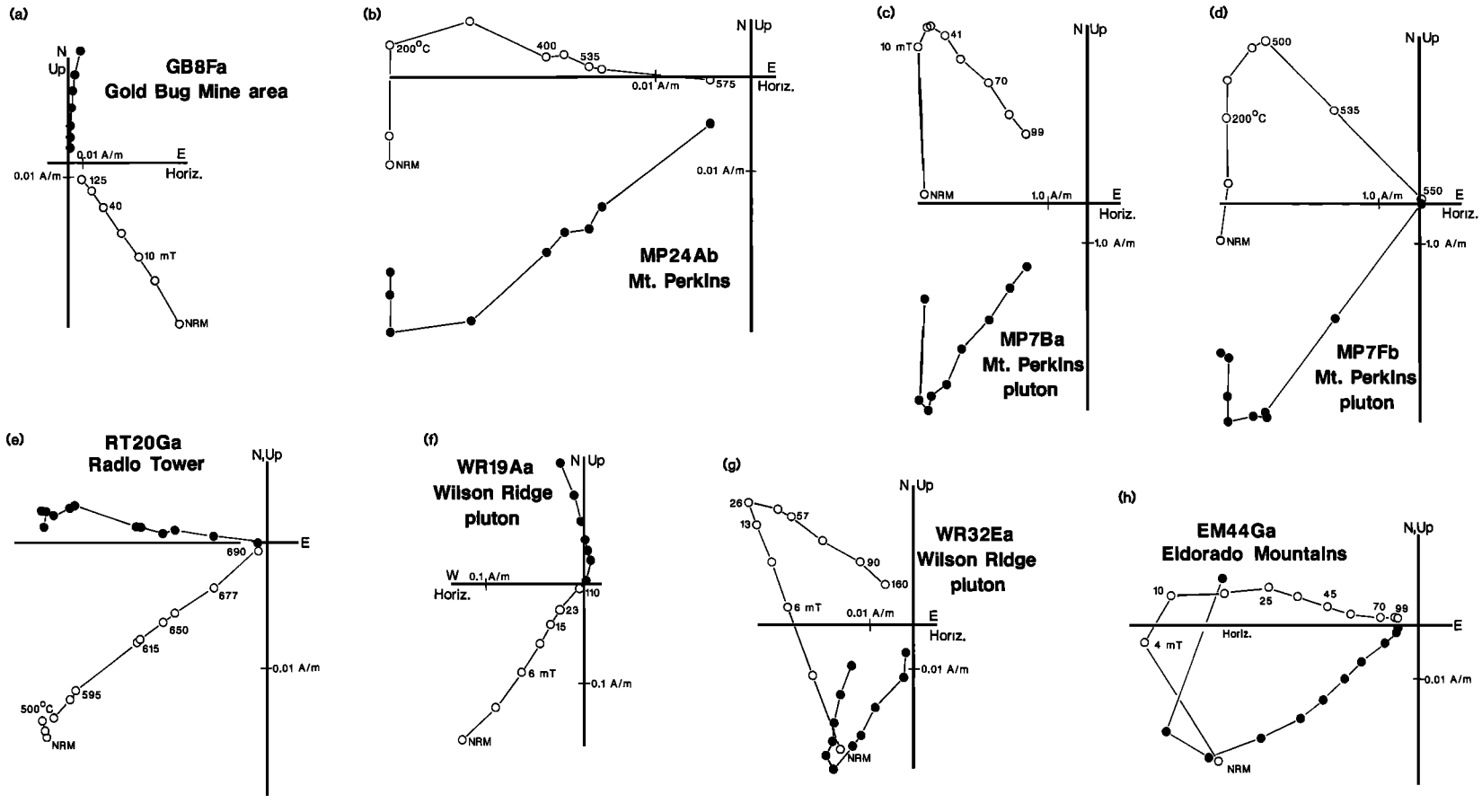



Fig. 8. Representative orthogonal demagnetization diagrams from crystalline rocks. Conventions follow as in Figure 5. (a) AF demagnetization, Miocene (?) dike, Gold Bug Mine area, axis of accommodation zone, central Black Mountains, site GB8; ChRM: 5/53, MAD = 0.9°. (b) Thermal demagnetization, Proterozoic gneiss, Mount Perkins block, central Black Mountains, site MP24; ChRM: 232/-7. (c) AF demagnetization, 16.8 +/- 0.4 Ma Mount Perkins pluton, Mount Perkins block, central Black Mountains, site MP7; ChRM: 216/-34. (d) Thermal demagnetization, 16.8 +/- 0.4 Ma Mount Perkins pluton, Mount Perkins block, central Black Mountains, site MP7; ChRM: 218/-30. (e) Thermal demagnetization, 73.3 +/- 1.5 Ma Radio Tower pluton, southern part of northern Black Mountains crystalline terrane, site RT20; ChRM: 279/37, MAD = 1.7°. (f) AF demagnetization, Wilson Ridge pluton, 13.5 +/- 0.4

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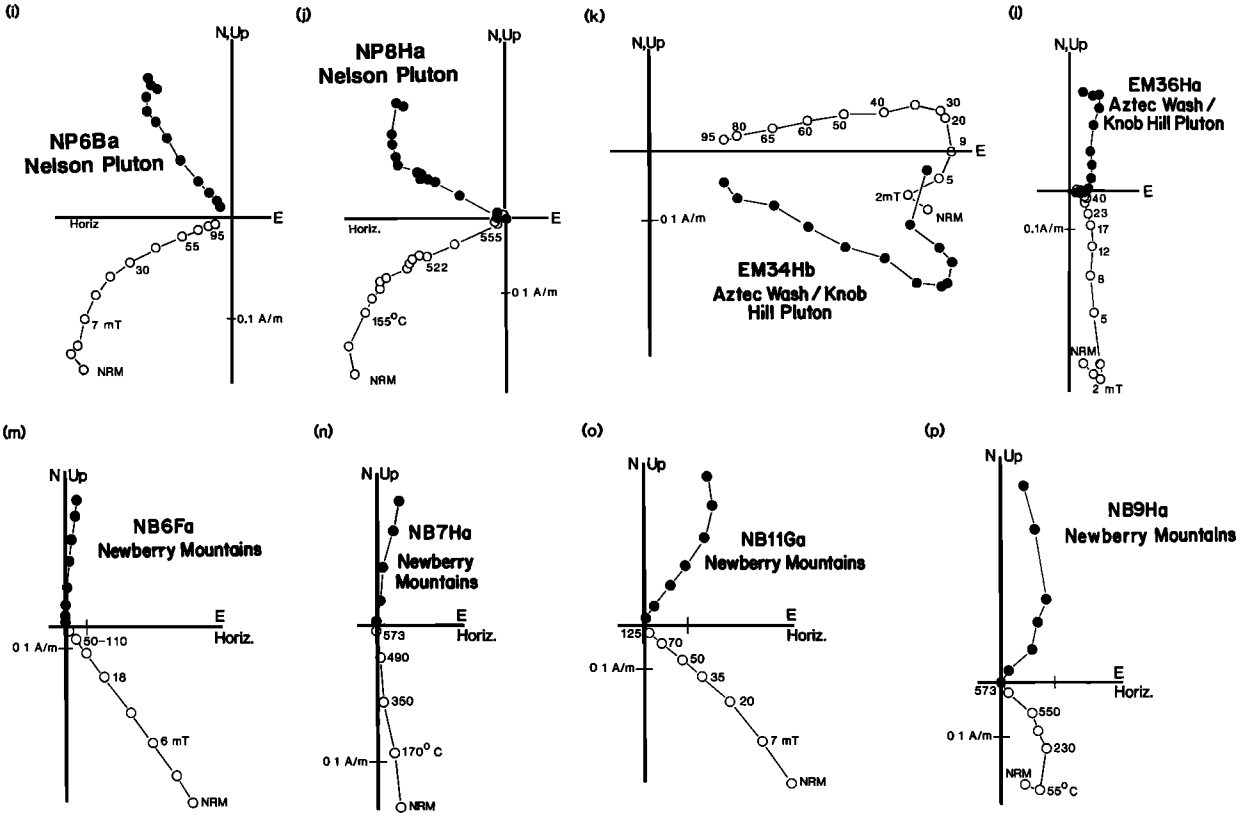



Fig. 8. (continued)

Ma quartz monzonite phase, northern part of northern Black Mountains crystalline terrane, site WR19; ChRM: 19/46, MAD = 4.2°. (g) AF demagnetization, Wilson Ridge pluton, 13.4 +/- 0.4 Ma diorite phase, northern part of northern Black Mountains crystalline terrane, site WR32; ChRM: 203/-44, MAD = 5.2°. (h) AF demagnetization, Miocene (?) dike, southwestern part of southern Eldorado Mountains crystalline terrane, site EM44; ChRM: 234/-11, MAD = 3.3°. (i) AF demagnetization, 16.9 +/- 0.5 Ma Nelson pluton, central Eldorado Mountains, site NP6; ChRM: 321/25, MAD = 2.1°. (j) Thermal demagnetization, 16.9 +/- 0.5 Ma Nelson pluton, Eldorado Mountains, site NP8; ChRM: 296/34, MAD = 8.3°. (k) AF demagnetization, 16.2 +/- 0.5 Ma Aztec Wash/Knob Hill pluton, northern part of southern Eldorado Mountains crystalline terrane, site EM34; ChRM: 116/-9, MAD = 3.2°. (l) AF demagnetization, 16.2 +/- 0.5 Ma Aztec Wash/Knob Hill pluton, northern part of southern Eldorado Mountains crystalline terrane, site EM36; ChRM (based on last few demagnetization steps): 108/10, MAD = 5.8°. (m) AF demagnetization, Miocene (?) mafic dike, central part of Newberry Mountains crystalline terrane, site NB6; ChRM: 3/53, MAD = 2.1°. (n) Thermal demagnetization, Miocene (?) intermediate dike, central part of Newberry Mountains crystalline terrane, site NB7; ChRM: 2/61, MAD = 2.6°. (o) AF demagnetization, Miocene (?) mafic dike, northern part of Newberry Mountains crystalline terrane, site NB11; ChRM: 37/34, MAD = 1.7°. (p) Thermal demagnetization, Proterozoic (?) pluton, northern part of Newberry Mountains crystalline terrane, site NB9; ChRM: 43/35, MAD = 4.3°.

(Figure 7b and Table 5). In the Mount Perkins block a north-south trending Tertiary nonconformity dips steeply (70°-90°+) to the west (Figure 4), supporting the interpretation of the paleomagnetic data from the crystalline rocks.

The Mockingbird Mine fault is a major east dipping low-angle normal fault on the east flank of the Mount Perkins block (Figure 4). Geologic and paleomagnetic data indicate significant tilting of both the hanging wall and footwall of the fault. This suggests that the fault nucleated at a steep dip and was rotated to its present gentle dip during block tilting.

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## Northern Black Mountains Crystalline Terrane

Directly north of the accommodation zone in the southern part of the northern Black Mountains crystalline terrane (Figure 4), the 73.3 +/- 1.5 Ma (Table 1, 88-175) Radio Tower pluton yielded a ChRM of westerly declination and positive inclination (Figure 7c and Table 5). $T_{ub}$ values of 400°-695°C (Figure 8e) indicate that the ChRM of the pluton resides in both titanomagnetite and hematite. The anomalous direction of the ChRM suggests that the hematite formed prior to tilting, perhaps during Late Cretaceous alteration and mineralization. The ChRM in the Radio Tower pluton implies 60° of east tilting about a N30°W axis.

On the basis of paleomagnetic and geologic data we propose that the southern part of the northern Black Mountains crystalline terrane is part of a highly tilted fault block rather than a flexed lower plate. The east tilted Radio Tower pluton is situated in the footwall of the gently west dipping Van Deemen Mine normal fault (Figure 4). Tertiary strata in the hanging wall of the Van Deemen Mine fault have an average attitude of N10°W, 60°E (Figure 9). The uniform tilts of the hanging wall and footwall suggest that the Van Deemen Mine fault nucleated at a high angle and was rotated to a gentle dip by Miocene block tilting and that the crystalline footwall is part of an upper plate fault block (Figure 2).

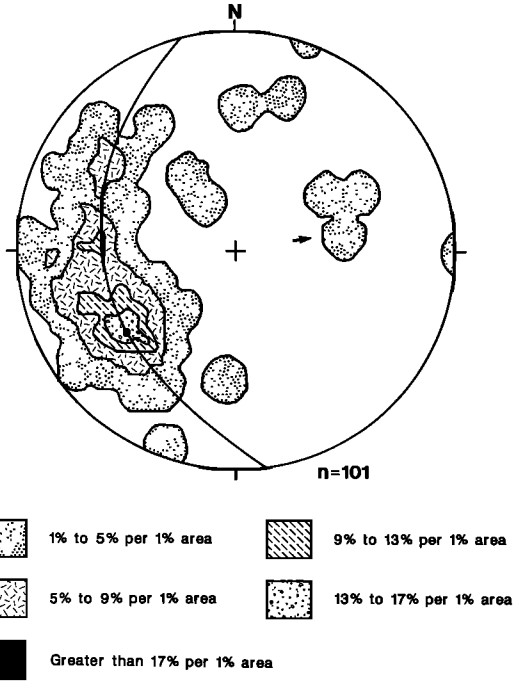



Fig. 9. Poles to layering of Miocene volcanic and sedimentary strata in the hanging wall of Van Deemen Mine fault, indicating an average attitude of N10°W, 60°NE. The great circle defined by the poles to layering may result from minor folding associated with corrugations in the Van Deemen Mine fault surface.

In the northern part of the northern Black Mountains crystalline terrane, 35 km north of the accommodation zone (Figure 4), a ChRM of northerly declination and moderate positive inclination was removed from the 13.5 +/- 0.4 Ma Wilson Ridge pluton [Anderson et al., 1972; Feuerbach and Smith, 1986; Weber and Smith, 1987; Larsen and Smith, 1990] and dikes of probable Miocene age (Figure 7d and Table 5). Median destructive inductions (Figures 8f and 8g) and $T_{ub}$ values indicate that the ChRM resides in fine-grained titanomagnetite. We interpret this ChRM as a TRM acquired during original intrusion and cooling.

The paleomagnetic data indicate negligible tilting of the northern part of the northern Black Mountains crystalline terrane. Detailed geologic mapping and petrologic characteristics also suggest negligible tilting of the Wilson Ridge pluton [Feuerbach, 1986; Naumann, 1987]. The paleomagnetic data are compatible with interpretations of the northern part of the northern Black Mountains crystalline terrane as the lower plate of a detachment terrane and the Saddle Island fault (Figure 4) as a major detachment fault [Weber and Smith, 1987; Duebendorfer et al., 1990]. ChRMs from the east and west sides of the northern part of the terrane are similar, indicating little flexing of these potential lower plate rocks.

## Southern Eldorado Mountains Crystalline Terrane

Hypabyssal dikes of probable Miocene age in the southwestern part of the southern Eldorado Mountains crystalline terrane (Figure 4) yielded a ChRM of southwesterly declination and negative inclination (Figure 7e and Table 5). Moderate to high median destructive inductions (Figure 8h) and $T_{ub}$ values indicate that the ChRM resides primarily in fine-grained titanomagnetite. These data suggest 45°-60° of west tilting about a north-south axis of the southern part of the terrane. A depositional contact between west tilted Tertiary strata and the southwestern part of the terrane [Longwell et al., 1965] and an eastward trend from hypabyssal dike swarms to coarser-grained plutonic rocks also support the hypothesis of significant west tilting.

In the northernmost part of the crystalline terrane, a ChRM of positive inclination and west-northwest declination was isolated from the 16.9 +/- 0.5 Ma Nelson pluton (Table 1, 88-180) (Figure 4). Median destructive inductions and $T_{ub}$ values (Figures 8i and 8j) indicate that fine-grained low-Ti titanomagnetite carries the ChRM. These

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data suggest that the Nelson pluton has been tilted 50°-60° to the east about a north-south axis (Figure 7f and Table 5). The Nelson pluton intrudes Tertiary strata tilted steeply eastward.

About 4 km south of the southernmost exposures of east tilted Tertiary strata in the Eldorado Mountains, a ChRM of east-southeast declination and shallow inclination (Figures 7g and 8k and Table 5) was isolated in the 16.2 +/- 0.5 Ma (Table 1, 88-179) Aztec Wash/Knob Hill pluton (Figure 4). These data imply 60°-90° of east tilting (or flexing) about a north-northwest trending axis.

Paleomagnetic data and geologic relations suggest that the entire southern Eldorado Mountains crystalline terrane may be part of oppositely tilted fault blocks rather than lower plate rocks of detachment terranes. The northern part of the crystalline terrane is tilted 50°-90° to the east, which is comparable to that of the southernmost exposures of east tilted Tertiary strata (Figure 4). The southern part of the terrane, which rests in the footwall of the gently east dipping Dupont Mountain fault, is tilted at least 45°-60° to the west. Tertiary strata directly above the Dupont Mountain fault are tilted approximately 65° to the west. Similar amounts of tilting in the hanging wall and footwall of the Dupont Mountain fault imply that the crystalline footwall is part of an upper plate fault block. The Dupont Mountain fault probably originated at a moderate to steep dip and was in turn rotated to a gentle dip by dominolike block tilting. The accommodation zone may bisect the central part of the crystalline terrane 5-10 km south of the Nelson fault (Figure 4).

### Newberry Mountains Crystalline Terrane

Demagnetization of plutons and dikes isolated two ChRMs in the Newberry Mountains crystalline terrane, which lies approximately 50 km south of the accommodation zone. In the central part of the terrane, Miocene (?) dikes and plutons yielded a ChRM of northerly declination and positive inclination (Figures 7h, 8m and 8n, and Table 5). In contrast, a ChRM of northeasterly declination and positive inclination (Figure 7h and Table 5) was isolated from moderately east dipping mafic dikes (Figure 8o), of probable Miocene age, and a 1.4 Ga granite (one site) in the northern part of the terrane (Figure 8p). The demagnetization data indicate that the ChRMs reside in fine-grained titanomagnetite.

These paleomagnetic data, albeit preliminary, may be interpreted in several ways. The ChRM in the northern Newberry Mountains suggests approximately 40° of west tilting or flexing about a north-south axis. In the central part of the terrane the ChRM may be a VPTRM acquired after tilting or flexing, perhaps during uplift and erosion of either a tilted fault block or flexed lower plate. Alternatively, it may be a pretilt TRM, in which case a major structure may separate the northern and central Newberry Mountains. The northern area may be part of a moderately west tilted upper plate fault block or flexed lower plate, whereas the central part may rest in an unflexed lower plate setting.

## DISCUSSION

### Paleomagnetic Results

In the northern Colorado River extensional corridor and elsewhere, uncertainty in the timing of magnetization acquisition greatly limits paleomagnetic assessment of the magnitude of tilting of crystalline terranes. Remanence blocking in plutons may significantly postdate cooling below the closure temperatures of isotopic systems used to constrain age of emplacement. Thus the magnitude of tilting inferred from paleomagnetic data may commonly be a minimum estimate.

For example, paleomagnetic estimates of the magnitude of tilting are not entirely compatible with K/Ar isotopic age determinations for both the Mount Perkins and Nelson plutons (Table 1, 88-181 and 88-180). The 40°-50° of west tilting of the Mount Perkins pluton is approximately half that of the lower part of the Tertiary section in the Mount Perkins block. This suggests that the Mount Perkins pluton was emplaced during tilting. However, the 16.8 +/- 0.4 Ma K/Ar age determination on biotite indicates that emplacement and cooling of the pluton to approximately 300°C predated the onset of major extension, at approximately 16.4 Ma. In the northern Eldorado Mountains the Nelson pluton intrudes the lower part of the Tertiary section, which is tilted steeply eastward and locally overturned. Anderson et al. [1972] constrained most of the tilting in the northern Eldorado Mountains to between 15 and 13 Ma. As with the Mount Perkins pluton, the 16.9 +/- 0.5 Ma K/Ar isotopic age determination from the Nelson pluton is not compatible with only 50°-60° of tilting inferred from the paleomagnetic data. The Nelson and Mount Perkins plutons may have acquired their ChRMs after emplacement during uplift and cooling induced by Miocene extension. Alternatively, the plutons may have experienced changes in K/Ar systematics during postemplacement argillic or potassic alteration. The Nelson pluton is altered in most areas, as evidenced by highly sericitized plagioclase and chloritized mafic phases. It was emplaced along the east-west trending Nelson fault zone (Figure 4), which is characterized by significant hydrothermal alteration. In addition, the low K content of biotite suggests slight chloritization in the Mount Perkins pluton (Table 1, 88-181).

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Interpretation of data from Proterozoic gneiss in the Mount Perkins block is also complicated by uncertainties in the age of the ChRM. Acquisition of a VPTRM in Proterozoic gneiss more than 1 km from the margin of the Mount Perkins pluton may have coincided with uplift associated with Mesozoic/early Tertiary contraction and/or Miocene extension. The timing of uplift associated with erosional stripping of Paleozoic and Mesozoic strata in the northern Colorado River extensional corridor and, possibly, of VPTRM acquisition in the gneiss is poorly constrained but must predate the inception of widespread volcanism at approximately 20 Ma [Anderson et al., 1972]. This period of uplift may be related to Mesozoic thrusting farther to the west [Burchfiel et al., 1974; Burchfiel and Davis, 1988; Fleck and Carr, 1990] and deposition of southwesterly derived, early Tertiary gravels on the margin of the Colorado Plateau to the east [Young, 1982]. Magmatism, tectonic denudation, and uplift during Miocene time may have also induced acquisition of a VPTRM in the Proterozoic gneiss.

Paleomagnetic data can afford minimum estimates of the magnitude of tilting of a crystalline terrane but cannot be independently used to assess the relative timing of intrusion, magnetization acquisition, and regional deformation. In the northern Colorado River extensional corridor the magnitude of tilting of crystalline rocks, as inferred from paleomagnetic data, provides requisite, albeit incomplete, information for palinspastic reconstructions and estimates of extension.

## Regional Implications

Paleomagnetic data from volcanic rocks in the central Black Mountains imply little vertical axis rotation in the vicinity of the accommodation zone. These data, together with geologic mapping and structural analysis [Faulds, 1989; Faulds et al., 1990], indicate that the accommodation zone facilitated neither distributed strike-slip displacement nor significant relative movement between opposing tilt block domains. Vertical axis rotations are probably negligible in all localities sampled for this study, as evidenced by strong similarities in the Miocene structural grain between the sampled areas and central Black Mountains.

Paleomagnetic data from crystalline rocks (Figure 10 and Table 6) in the southern Eldorado and northern Black Mountains suggest that large portions of both crystalline terranes underwent significant horizontal axis rotation and probably correspond to parts of highly tilted fault blocks rather than to flexed lower plate rocks of a detachment terrane. Similar magnitudes of tilting of the hanging walls and footwalls of the Dupont Mountain and Van Deemen Mine

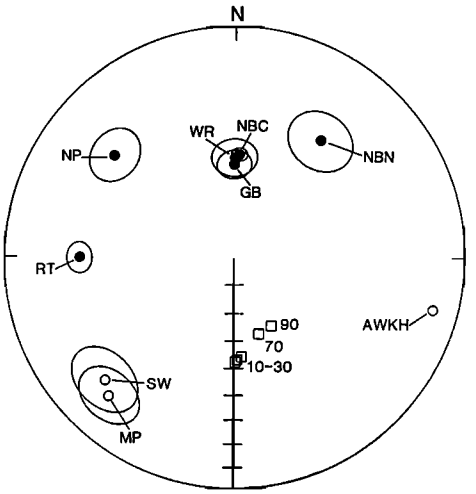



Fig. 10. Equal-area projection of in situ, locality site mean magnetizations from crystalline terranes, northern Colorado River extensional corridor. Conventions follow as in Figure 7. AWKH, 16.2 Ma Aztec Wash/Knob Hill pluton, northern part of southern Eldorado Mountains; GB, Miocene (?) dikes, axis of accommodation zone, central Black Mountains; MP, Mount Perkins block, central Black Mountains, all rock types included in calculation of group site mean; NBC, central Newberry Mountains; NBN, northern Newberry Mountains; NP, 16.9 Ma Nelson pluton, central Eldorado Mountains; RT, 73.3 Ma Radio Tower pluton, southern part of northern Black Mountains; SW, Miocene (?) dikes, southwestern part of southern Eldorado Mountains; WR, 13.5 Ma Wilson Ridge pluton, northern part of northern Black Mountains.

faults imply that tilting of the crystalline footwall rocks did not result from either excisement-incisement structures, which may transfer the lower plate of an early detachment fault into the upper plate of a subsequent detachment [e.g., Davis and Lister, 1988], or from flexing of the tectonically denuded footwall of a major detachment fault [Wernicke and Axen, 1988; Buck, 1988; Reynolds and Lister, 1990]. These processes would presumably produce differential tilting between much of the footwall and hanging wall of a detachment fault and possibly opposing senses of tilt in parts of an individual crystalline terrane (i.e., flexing). The available paleomagnetic data do not indicate flexing parallel to the extension direction in either the southern Eldorado or northern Black Mountains crystalline terranes.

Nevertheless, an interpretation depicting the entire southern Eldorado and northern Black Mountains crystalline terranes as large north-northwest trending antiforms in a flexed lower plate of their respective detachment

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TABLE 6. Paleomagnetic Data, Crystalline Rocks, In Situ, Group Site Means

| Locality | Site | Crystalline Unit | $N/N_0$ | D | I | $a_{95}$ | k |
|---|---|---|---|---|---|---|---|
| Central Black Mountains | | | | | | | |
| Gold Bug Mine area | GB | Miocene (?) mafic and intermediate dikes | 7/8 | 359.6 | 57.2 | 5.6 | 117 |
| Mount Perkins block | MP | Miocene dikes, 16.9 Ma granodiorite pluton, Proterozoic gneiss | 12/12 | 221.7 | -19.8 | 10.4 | 20 |
| Northern Black Mountains | | | | | | | |
| Radio Tower pluton (south part of terrane) | RT | 73.3 Ma granite pluton | 10/10 | 269.7 | 33.3 | 5.1 | 90 |
| Wilson Ridge area (north part of terrane) | WR | Miocene dikes and 13.5 Ma pluton | 13/13 | 359.8 | 54.8 | 7.6 | 26 |
| Southern Eldorado Mountains | | | | | | | |
| Southwest part of terrane | SW | Miocene (?) dike swarm | 5/5 | 225.8 | -24.2 | 14.8 | 28 |
| Nelson pluton (north part of terrane) | NP | 16.8 Ma granodiorite | 7/7 | 309.7 | 31.9 | 8.9 | 48 |
| Aztec Wash/Knob Hill pluton (north part of terrane) | AWKH | 16.2 Ma granodiorite | 3/3 | 105.1 | -12.1 | 43.7 | 9 |
| Newberry Mountains | | | | | | | |
| Central part of terrane | NBC | Miocene dikes and plutons | 6/6 | 2.2 | 53.6 | 2.3 | 599 |
| North part of terrane | NBN | Miocene mafic dikes and Proterozoic pluton | 3/3 | 36.4 | 37.3 | 10.5 | 139 |

Site abbreviations refer to those shown in Figures 4 and 11. Other notation same as in Table 2.

terranes cannot be ruled out. In the southern Eldorado Mountains the hingeline of the antiform would lie between the west tilted felsic dike swarm in the southwestern part of the terrane and east tilted Aztec Wash/Knob Hill pluton in the northeastern part (Figure 4). In the northern Black Mountains the negligibly tilted Wilson Ridge pluton may lie near the hingeline of a hypothetical antiform. This interpretation seems unlikely, however, for the southern Eldorado and southern part of the northern Black Mountains crystalline terranes in light of strong similarities between the inferred amount of tilting of crystalline rocks and that of Tertiary strata in nearby fault blocks. We therefore suggest that lower plate rocks do not crop out within the accommodation zone.

Geologic [Davis and Lister, 1988; Spencer and Reynolds, 1989; Duebendorfer et al., 1990], petrologic [Feuerbach, 1986; Naumann, 1987], and paleomagnetic data do indicate, however, that lower plate rocks may surface in both the east and west tilted domains to the north and south of the accommodation zone. The nature and location of the northern and southern terminations of lower plate rocks and the possible explanation for the apparent paucity of lower plate exposures within the accommodation zone warrant further discussion.

The northern and southern parts of the northern Black Mountains crystalline terrane appear to occupy the lower and upper plate, respectively, of a west dipping system of detachment faults. If the timing of major extension was contemporaneous in the northern and central Black Mountains, a major detachment must cut crystalline terrane between the Wilson Ridge and Van Deemen Mine areas. Geologic mapping [Anderson, 1978; Faulds, 1989] and reconnaissance have not identified such a structure.

However, major extension was apparently diachronous in the northern and central Black Mountains. More than 60% of the total magnitude of east tilting occurred between 15.9 +/- 0.4 Ma and 13.7 +/- 0.3 Ma (Table 1, 88-067 and 88-176) in the central Black Mountains [Faulds et al., 1990], whereas most of the tilting in the western Lake Mead area and northernmost Black Mountains took place between approximately 13 and 9 Ma [Duebendorfer and Wallin, 1991].

This diachroneity in the timing of extension may account for the seemingly contradictory positions of the northern and southern parts of the northern Black Mountains crystalline terrane within the system of west dipping detachment faults. In the earlier pulse of extension much of the northern Black Mountains crystalline terrane may have been situated in the upper plate of a west dipping detachment system and thus significantly tilted. Approximately 60° of east tilting of the southernmost part of the northern Black Mountains crystalline terrane (Figure 4), as inferred with paleomagnetic data from the 73.3 +/- 1.5 Ma Radio Tower pluton, probably occurred during this early period of extension. In the vicinity of the Black Mountains, tilting associated

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with the early west dipping detachment system and possibly the detachment system itself died out northward toward the western Lake Mead area (Figure 11) and also southward within the accommodation zone. The magnitude of extension in the early event apparently was not sufficient to bring lower plate crystalline rocks to the surface, at least not in the vicinity of the accommodation zone. In the second pulse of extension (13-9 Ma) much of the northern Black Mountains crystalline terrane, including the 13.5 Ma Wilson Ridge pluton, probably resided in the lower plate of a west dipping detachment

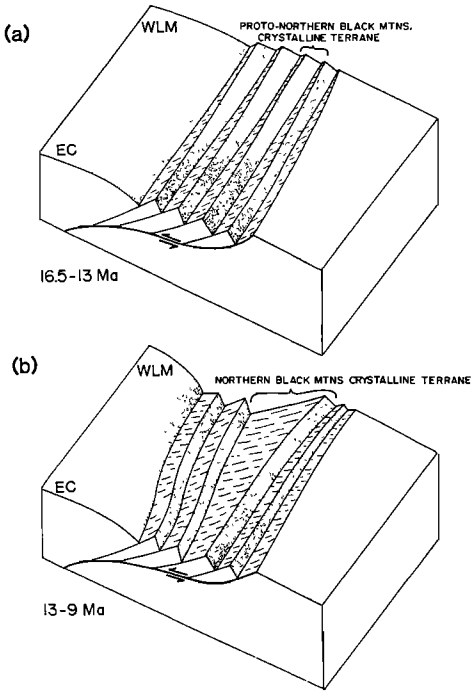



Fig. 11. Schematic block diagrams showing temporal patterns of extension in the east tilted domain. EC, Eldorado Canyon area, Arizona; WLM, western Lake Mead area. (a) Most of the tilting near the accommodation zone occurred between approximately 16.4 and 13.1 Ma. In the vicinity of the Black Mountains, displacement along this early system of west dipping normal faults and attendant east tilting probably died out northward toward the western Lake Mead area. (b) Major tilting in the western Lake Mead area took place between about 13 and 9 Ma [Duebendorfer and Wallin, 1991]. Part of the original upper plate in the early episode may have been transferred into a lower plate position during the later event. The 13.5 Ma Wilson Ridge pluton, which does not appear to be significantly tilted, apparently lies within an unflexed portion of the 13-9 Ma lower plate. In both events, the west dipping normal fault system terminated southward in the vicinity of the accommodation zone.

fault [Weber and Smith, 1987]. The paleomagnetic data from the Wilson Ridge pluton suggest that most of the northern Black Mountains crystalline terrane has not been appreciably flexed or tilted since 13.5 Ma. The upper plate of the early west dipping detachment system, in the vicinity of the Black Mountains, may have been excised [e.g., Davis and Lister, 1988] or transferred into the lower plate of the later system. The west dipping detachment system in the second event also terminated southward in the accommodation zone.

The cumulative effect of both events is that the exposed structural level, with respect to Tertiary extension, deepens and areal extent of lower plate rocks increases northward, away from the accommodation zone (Figure 12). The northward increase in depth of exposure is well illustrated by the distribution of Tertiary mylonites. Tertiary mylonites do not crop out near the accommodation zone but do surface about 50 km north of the zone in the western Lake Mead area at Saddle Island [Duebendorfer et al., 1990] and along the western flank of the South Virgin Mountains (J. E. Fryxell et al., Gold Butte crustal section, South Virgin Mountains, Nevada, submitted to Tectonics, 1990) (Figure 4).

Similarly, the exposed structural level deepens and areal extent of lower plate rocks increases southward, away from the accommodation zone, within the west tilted domain (Figure 12). Paleomagnetic data and geologic relations suggest that highly tilted, upper plate fault blocks constitute the entire southern Eldorado Mountains crystalline terrane. However, lower plate rocks appear to surface farther south in the west tilted domain, as evidenced by the large lateral extent, in the direction of displacement, of some crystalline terranes (e.g., Whipple Mountains) [Davis et al., 1980, 1986; Davis and Lister, 1988], P-T-t paths of crystalline rocks [Anderson et al., 1988], and abundance of Tertiary mylonitic rocks [Spencer and Reynolds, 1989]. With only minor interruptions, lower-plate rocks may continue northward from the Whipple Mountains into the Chemehuevi, Sacramento, Dead, and Newberry mountains [Spencer, 1985; John, 1987; Howard and John, 1987; Spencer and Reynolds, 1989]. The apparently untilted central part of the Newberry Mountains crystalline terrane may mark the northernmost extent of lower plate rocks in the west tilted domain (Figure 12). Alternatively, the entire Newberry Mountains terrane may correspond to the lower part of a highly tilted fault block and to the northernmost extent, in the west tilted domain, of rocks dominated by an uplift-related, essentially posttilt VPTRM. The northward progression of deformation in the west tilted domain during early to middle Miocene time implies, however, that

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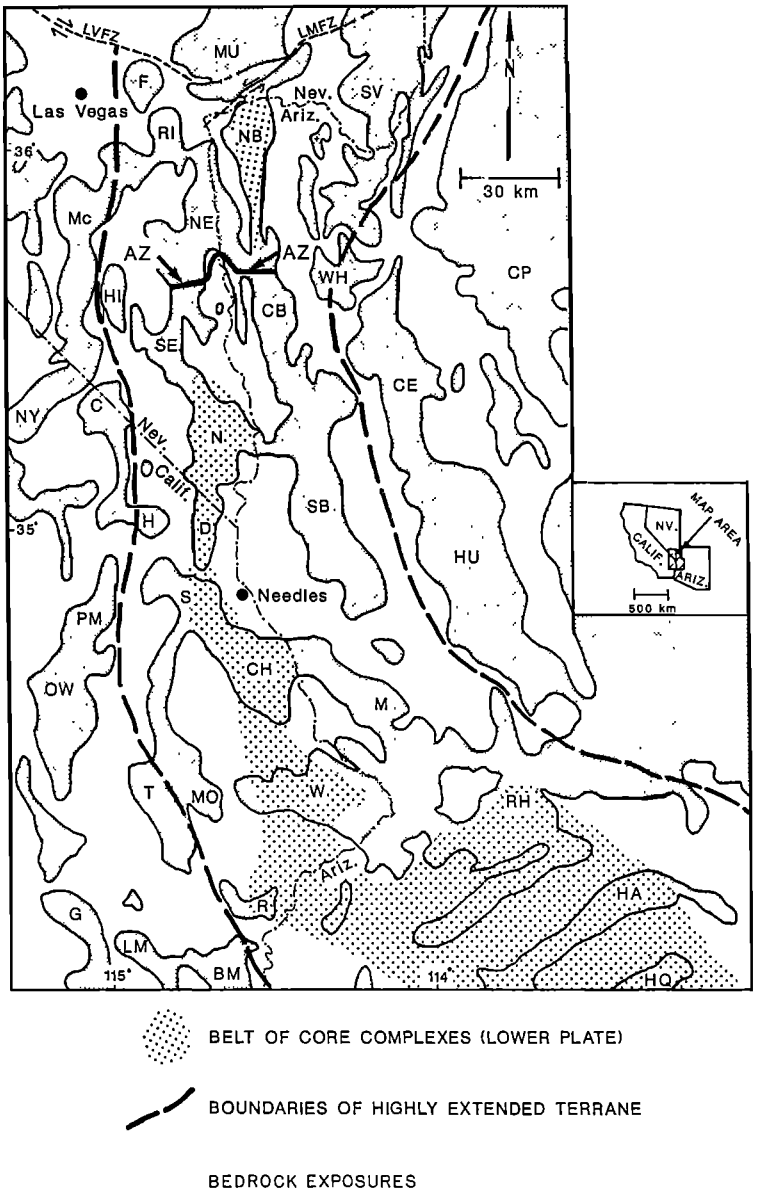



Fig. 12. Distribution of possible lower plate rocks, western Arizona, southern Nevada, and southeastern California; modified from Spencer and Reynolds [1989]. Exposures of lower plate rocks and width of highly extended terrane appear to decrease toward the accommodation zone in the northern Colorado River extensional corridor. AZ, accommodation zone; BM, Big Maria Mountains; C, Castle Mountains; CB, central Black Mountains; CE, Cerbat Mountains; CH, Chemehuevi Mountains; CP, Colorado Plateau; D, Dead Mountains; F, Frenchman Mountain; G, Granite Mountains; H, Homer Mountain; HA, Harcuvar Mountains; HQ, Harquahala Mountains; HI, Highland Spring Range; HU, Hualapai Mountains; LVFZ, Las Vegas Valley fault zone; LM, Little Maria Mountains; LMFZ, Lake Mead fault zone; M, Mohave Mountains; Mc, McCullough Range; MO, Mopah Mountains; MU, Muddy Mountains; N, Newberry Mountains; NB, northern Black Mountains; NE, northern Eldorado Mountains; NY, New York Mountains; OW, Old Woman Mountains; PM, Piute Mountains; R, Riverside Mountains; RH, Rawhide Mountains; RI, River Mountains; S, Sacramento Mountains; SB, southern Black Mountains; SE, southern Eldorado Mountains; SV, South Virgin Mountains; T, Turtle Mountains; W, Whipple Mountains; WH, White Hills.

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excisement/incisement processes may also complicate lower and upper plate relations. In any case, the depth of exposure appears to progressively decrease northward between the Whipple and southern Eldorado mountains.

East dipping detachment faults may project beneath the surface near the northern end of the Newberry Mountains and lie at relatively shallow levels beneath the southern part of the southern Eldorado Mountains crystalline terrane. The northernmost expression of the east dipping detachment system is probably the east dipping Dupont Mountain fault. Although decreasing significantly northward, displacement on the Dupont Mountain fault was sufficient to produce a narrow zone of west tilted fault blocks, which extends 8 km farther north than other parts of the west tilted domain (Figure 4).

A northward decrease in offset on the east dipping detachment system between the Whipple and southern Eldorado mountains presumably accounts for the commensurate decrease in both the exposed structural level, with respect to Tertiary extension, and abundance of mylonitic footwall rocks. The possibility that slip on the Whipple Mountains detachment system is transferred to slip on a series of lesser normal faults, as demonstrated by Richard et al. [1990] for the southeastern edge of the west tilted domain, seems less likely in the northern part of the west tilted domain, as the width of highly extended terrane also decreases appreciably to the north (Figure 12). In addition, the intermeshing of upper plate, conjugate normal faults and possibly conjugate detachment faults may have prevented accumulation of sufficient displacement along any single detachment to completely denude lower plate rocks near the accommodation zone. Accordingly, mylonitic lower plate rocks may rarely crop out in rifted continental crust in the vicinity of accommodation zones.

The distribution of lower plate rocks and tapering of highly extended terrane toward the accommodation zone (Figure 12) imply that the magnitude of upper crustal extension in the Colorado River extensional corridor progressively decreases toward the zone in both the west and east tilted domains. Greater extension in nearby regions may, in general, compensate for lesser amounts of extension near the accommodation zone. The magnitude of extension also decreases toward major accommodation zones in the Lake Tanganyika region of the east African rift system (Morley, 1988).

The apparent decrease in the magnitude of upper crustal extension toward some accommodation zones may imply that opposing detachment systems propagate toward and die out in the zones [e.g., Bosworth, 1985; Bosworth et al., 1986]. In the Colorado River extensional corridor, major extension does young northward in the west tilted domain from early to middle Miocene time [Howard and John, 1987; Anderson et al., 1972; Faulds et al., 1990]. However, a complementary trend has not been recognized in the east tilted domain. Major extension in the east tilted domain appears to young northward away from the accommodation zone rather than toward the zone.

We therefore suggest that the accommodation zone in the northern Colorado River extensional corridor served as a long-lived rupture barrier [e.g., Aki, 1979; King, 1986] that facilitated both temporal and geometric changes in extension throughout Miocene time rather than a relatively short-term feature that accommodated a dovetailing of laterally propagating detachment systems of opposite dip. Many Quaternary accommodation zones in the Great Basin also correspond to long-lived rupture barriers that originally formed during Tertiary time [Fonseca, 1988; Thenhaus and Barnhard, 1989].

## CONCLUSIONS

Paleomagnetic data provide important constraints on the structural development of a Miocene accommodation zone in the northern Colorado River extensional corridor. Data from Miocene volcanic rocks indicate negligible vertical axis rotation within the zone. This conclusion supports previous interpretations [Faulds et al., 1990] that the zone neither facilitated distributed strike-slip displacement at exposed structural levels nor significant relative movement between opposing tilt block domains. Paleomagnetic data from crystalline rocks and geologic relations suggest that lower plate rocks do not crop out near the zone but may surface in both the east and west tilted domains 35-50 km north and south of the zone. The apparent distribution of lower plate rocks and Tertiary mylonites suggests that structural levels, with respect to Tertiary extension, shoal toward the accommodation zone. The shallowing of structural levels and tapering of the extensional corridor toward the zone further imply that the magnitude of upper crustal extension may decrease toward the zone. Temporal patterns of major extension within the extensional corridor suggest that major detachment faults did not propagate toward the accommodation zone. Instead, the accommodation zone probably served as a persistent rupture barrier [e.g., Aki, 1979; King, 1986] between conjugate normal fault systems from middle to late Miocene time.

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## REFERENCES


Aki, K., Characterization of barriers on an earthquake fault, J. Geophys. Res., 84, 6140-6148, 1979.

Anderson, J. L., A. P. Barth, and E. D. Young, Mid-crustal Cretaceous roots of Cordilleran metamorphic core complexes, Geology, 16, 366-369, 1988.

Anderson, R. E., Thin skin distension in Tertiary rocks of southeastern Nevada, Geol. Soc. Am. Bull., 82, 43-58, 1971.

Anderson, R. E., Geologic map of the Black Canyon 15-minute quadrangle, Mohave County, Arizona, and Clark County, Nevada, U.S. Geol. Surv. Geol. Quad. Map, GQ-1394, 1978.

Anderson, R. E., C. R. Longwell, R. L. Armstrong, and R. F. Marvin, Significance of K-Ar ages of Tertiary rocks from the Lake Mead region, Nevada-Arizona, Geol. Soc. Am. Bull., 83, 273-288, 1972.

Bartley, J. M., J. M. Fletcher, and A. F. Glazner, Tertiary extension and contraction of lower-plate rocks in the central Mojave metamorphic core complex, southern California, Tectonics, 9, 521-534, 1990.

Beck, M. E., Jr., Paleomagnetic record of plate margin tectonic processes along the western edge of North America, J. Geophys. Res., 85, 7115-7131, 1980.

Bosworth, W., Geometry of propagating continental rifts, Nature, 316, 625-627, 1985.

Bosworth, W., J. Lambiase, and R. Keisler, A new look at Gregory's rift: The structural style of continental rifting, Eos Trans. AGU, 67, 577-583, 1986.

Buck, W. R., Flexural rotation of normal faults, Tectonics, 7, 959-973, 1988.

Burchfiel, B. C., and G. A. Davis, Mesozoic thrust faults and Cenozoic low-angle normal faults, eastern Spring Mountains, Nevada, thrust complex, Nevada, and Clark Mountains thrust complex, California, This Extended Land, Geological Journeys in the Southern Basin and Range, Field Trip Guidebook, edited by D.L. Weide and M.L. Faber, pp. 87-106, Cordilleran Section, Geological Society of America, Las Vegas, Nev., 1988.

Burchfiel, B. C., R. J. Fleck, D. T. Secor, R. R. Vicelette, and G. A. Davis, Geology of the Spring Mountains, Nevada, Geol. Soc. Am. Bull., 85, 1013-1022, 1974.

Butler, R. F., G. E. Gehrels, W. C. McClelland, S. R. May, and D. Klepacki, Discordant paleomagnetic poles from the Canadian Coast Plutonic Complex: Regional tilt rather than large-scale displacement?, Geology, 17, 691-694, 1989.

Chamberlin, R. M., Structural development of the Lemitar Mountains, an intrarift tilted fault block uplift, central New Mexico (abstract), in International Symposium on the Rio Grande Rift, Los Alamos Sci. Lab. Conf. Proc. LA-7487-C, pp. 22-24, Santa Fe, N. M., 1978.

Chamberlin, R. M., Cenozoic domino-style crustal extension in the Lemitar Mountains, New Mexico: A summary, in Socorro Region II, edited by C. E. Chapin and J. F. Callender, Field Conf. Guideb., N. M. Geol. Soc. Guidebook, 34, 111-118, 1983.

Chan, L. S., Apparent tectonic rotations, declination anomaly equations, and declination anomaly charts, J. Geophys. Res., 93, 12151-12158, 1988.

Crittenden, M. D., Jr., P. J. Coney, and G. H. Davis (Eds.), Cordilleran Metamorphic Core Complexes, 490 pp., Mem. Geol. Soc. Am., 153, 1980.

Dagley, P., and J. D. A. Piper, The application of palaeomagnetism to extensional tectonics: A palaeomagnetic study of the Parker district, Basin and Range province, Arizona, Paleomagnetic Rotations and Continental Deformation, edited by C. Kissel and C. Laj, pp. 293-311, Kluwer Academic Boston, Mass., 1989.

Davis, G. A., and G. S. Lister, Detachment faulting in continental extension: Perspectives from the southwestern U. S. Cordillera, Spec. Pap. Geol. Soc. Am., 218, 133-159, 1988.

Davis, G. A., J. L. Anderson, E. G. Frost, and T. J. Shackelford, Mylonitization and detachment faulting in the Whipple-Buckskin-Rawhide Mountains terrane, southeastern California and western Arizona, edited by M. D. Crittenden, Jr., P. J. Coney, and G. H. Davis, Mem. Geol. Soc. Am., 153, 79-129, 1980.

Davis, G. A., G. S. Lister, and S. J. Reynolds, Structural evolution of the Whipple and South Mountains shear zones, southwestern United States, Geology, 14, 7-10, 1986.

Davis, G. H., Shear-zone model for the origin of metamorphic core complexes, Geology, 11, 342-347, 1983.

Davis, G. H., A shear-zone model for the structural evolution of metamorphic core complexes in southeastern Arizona, Geol. Soc. Spec. Pub. London, 28, 247-266, 1987.

Demarest, H. H., Error analysis for the determination of tectonic rotation from paleomagnetic data, J. Geophys. Res., 88, 4321-4328, 1983.

Duebendorfer, E. M., and E. T. Wallin, Basin development and syntectonic sedimentation associated with kinematically coupled strike-slip and detachment faulting, southern Nevada, Geology, 19, 87-90, 1991.

Duebendorfer, E. M., A. J. Sewall, and E. I. Smith, The Saddle Island detachment: An evolving shear zone in the Lake Mead area, Nevada, in Basin and Range Extensional Tectonics Near the Latitude of Las Vegas, Nevada, edited by B. Wernicke, Mem. Geol. Soc. Am., 176, 77-97, 1990.

Ebinger, C. J., Geometric and kinematic development of border faults and accommodation zones, Kivu-Rusizi rift, Africa, Tectonics, 8, 117-133, 1989.

Faulds, J. E., Structural development of a major extensional accommodation zone in the Basin and Range province, northwestern Arizona and southern Nevada: Implications for kinematic models of continental extension, Ph.D. dissertation, 263 pp., Univ. of New Mexico, Albuquerque, 1989.

Faulds, J. E., J. W. Geissman, and C. K. Mawer, Anisotropy of magnetic susceptibility data from the Mt. Perkins pluton, central Black Mountains, Arizona: Implications for evaluation of extension directions, Eos Trans. AGU, 68, 1251, 1987.

Faulds, J. E., F. L. Hillemeyer, and E. I. Smith, Geometry and kinematics of a Miocene "accommodation zone" in the central Black and southern Eldorado Mountains, Arizona and Nevada, This Extended Land, Geological Journeys in the Southern Basin and Range, Field Trip Guidebook, edited by D.L. Weide and M.L. Faber, pp. 293-310, Cordilleran Section, Geological Society of America, Las Vegas, Nev., 1988a.

Faulds, J. E., J. W. Geissman, and C. K. Mawer, Paleomagnetic evidence of large-magnitude tilting of crystalline terranes in the northern Colorado River extensional corridor, northwestern Arizona and southern Nevada, Eos Trans. AGU, 69, 1163-1164, 1988b.

Faulds, J. E., J. W. Geissman, and C. K. Mawer, Structural development of a major extensional accommodation zone in the Basin and Range province, northwestern Arizona and southern Nevada: Implications for kinematic models of continental extension, in Basin and Range Extensional Tectonics Near the Latitude of Las Vegas, Nevada, edited by B. Wernicke, Mem. Geol. Soc. Am., 176, 37-76, 1990.

Feuerbach, D. L., Geology of the Wilson Ridge pluton; a mid-Miocene quartz monzonite intrusion in the northern Black Mountains, Mohave County, Arizona and Clark County, Nevada, M.S. thesis, 79 pp., Univ. of Nevada, Las Vegas, 1986.

Feuerbach, D. L., and E. I. Smith, The mid-Miocene Wilson Ridge pluton: A subvolcanic intrusion in the Lake Mead region, Arizona and Nevada, Eos Trans. AGU, 67, 1262, 1986.

Fisher, R. A., Dispersion on a sphere, Proc. R. Soc. London Ser. A, 217, 295-305, 1953.

Fleck, R. J., and M. D. Carr, The age of the Keystone thrust: Laser-fusion $^{40}Ar/^{39}Ar$ dating of foreland basin deposits, southern Spring Mountains, Nevada, Tectonics, 9, 467-476, 1990.

Fonseca, J., The Sou Hills: A barrier to faulting in the central Nevada seismic belt, J. Geophys. Res., 93, 475-489, 1988.

Frost, E. G., and D. L. Martin (Eds.), Mesozoic-Cenozoic Tectonic Evolution of the Colorado River Region, California, Arizona, and Nevada, 608 pp., Cordilleran Publishers, San Diego, Calif., 1982.

Geissman, J. W., Paleomagnetism and late Cenozoic strike-slip tectonism in the Hoover Dam area, southeast extension of the Lake Mead fault zone, Nevada and Arizona, Eos Trans. AGU, 67, 922, 1986.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, J. Geophys. Res., 54, 131-167, 1949.

Howard, K. A., and B. E. John, Crustal extension along a rooted system of imbricate low-angle faults: Colorado River extensional corridor, California and Arizona, Geol. Soc. Spec. Publ. London, 28, 299-311, 1987.

Irving, E., and D. A. Archibald, Bathozonal tilt corrections to paleomagnetic data from mid-Cretaceous plutonic rocks: Examples from the Omineca Belt, British Columbia, J. Geophys. Res., 95, 4579-4585, 1990.

Irving, E., and G. A. Irving, Apparent polar wander paths: Carboniferous through Cenozoic and the assembly of Gondwana, Geophys. Surv., 5, 141-188, 1982.

John, B. E., Geometry and evolution of a mid-crustal extensional fault system: Chemehuevi Mountains, southeastern California, Geol. Soc. Spec. Publ. London, 28, 313-336, 1987.

King, G. C. P., Speculations on the geometry of the initiation and termination process of earthquake rupture and its relation to morphology and geological structure, Pure Appl. Geophys., 124, 567-585, 1986.

Kirschvink, J. L., The least squares line and plane and the analysis of paleomagnetic data: Geophys. J. R. Astron. Soc., 62, 699-718, 1980.

Larsen, L. L., and E. I. Smith, Mafic enclaves in the Wilson Ridge pluton, northwestern Arizona, Implications for the generation of a calc-alkaline intermediate pluton in an extensional environment, J. Geophys. Res., 17,693-17,716, 1990.

Lister, G. S., M. A. Etheridge, and P. A. Symonds, Detachment faulting and the evolution of passive continental margins, Geology, 14, 246-250, 1986.

Longwell, C. R., E. H. Pampeyan, B. Bowyer, and R. J. Roberts, Geology and mineral deposits of Clark County, Nevada, Bull. Nev. Bur. Mines Geol., 62, 218 pp., 1965.

MacDonald, W. D., Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies, J. Geophys. Res., 85, 3659-3669, 1980.

Mankinen, E. A., E. E. Larson, C. S. Gromme, M. Prevot, and R. S. Coe, The Steen Mountain (Oregon) geomagnetic polarity transition, 3, Its regional significance, J. Geophys. Res., 92, 8057-8076, 1987.

Marquis, G., and E. Irving, Observing tilts in midcrustal rocks by paleomagnetism: Examples from southeast British Columbia, Tectonics, 9, 925-934, 1990.

Mathis, R. S., Mid-Tertiary detachment faulting in the southeastern Newberry Mountains, Clark County, Nevada, in Mesozoic-Cenozoic Tectonic Evolution of the Colorado River Region, California, Arizona, and Nevada, edited by E. G. Frost and D. L. Martin, pp. 326-340, Cordilleran, San Diego, Calif., 1982.

May, S. R., M. E. Beck, and R. F. Butler, North American apparent polar wander, plate motion, and left-oblique convergence: Late Jurassic-early Cretaceous orogenic consequences, Tectonics, 8, 443-451, 1989.

McElhinny, M. W., Statistical significance of the fold test in paleomagnetism, Geophys. J. R. Astron. Soc., 8, 338-340, 1964.

Merrill, R. T., and M. W. McElhinny, The Earth's Magnetic Field, 401 pp.,


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Academic Press, San Diego, Calif., 1983.
Miller, E. L., P. B. Gans, and J. Garing, The Snake Range decollement: An exhumed mid-Tertiary ductile-brittle transition, Tectonics, 2, 239-263, 1983.
Morley, C. K., Variable extension in Lake Tanganyika, Tectonics, 7, 785-801, 1988.
Morton, W. H., and R. Black, Crustal attenuation in Afar, Afar Depression of Ethiopia, edited by A. Pilger and A. Roster, pp. 55-65, Inter-Union Commission on Geodynamics, E. Schweizerbartische Verlagsbuchhandlung, Stuttgart, Germany, 1975.
Naumann, T. R., Geology of the central Boulder Canyon quadrangle, Clark County, Nevada, M.S. thesis, 68 pp., Univ. of Nevada, Las Vegas, 1987.
Nelson, M. R., and C. H. Jones, Paleomagnetism and crustal rotations along a shear zone, Las Vegas, southern Nevada, Tectonics, 6, 13-33, 1987.
Potter, D. K., and A. Stephenson, Single-domain particles in rocks and magnetic fabric analysis, Geophys. Res. Lett., 15, 1097-1100, 1988.
Proffett, J. M., Jr., Cenozoic geology of the Yerington district, Nevada, and implications for the nature and origin of Basin and Range faulting, Geol. Soc. Am. Bull., 88, 247-266, 1977.
Pullaiah, G., E. Irving, K. L. Buchan, and D. J. Dunlop, Magnetization changes caused by burial and uplift, Earth Planet. Sci. Lett., 28, 133-143, 1975.
Reynolds, S. J., and G. S. Lister, Folding of mylonitic zones in Cordilleran metamorphic core complexes: Evidence from near the mylonitic front, Geology, 18, 216-219, 1990.
Reynolds, S. J., and J. E. Spencer, Evidence for large-scale transport on the Bullard detachment fault, west-central Arizona, Geology, 13, 353-356, 1985.
Richard, S. M., J. E. Fryxell, and J. F. Sutter, Tertiary structure and thermal history of the Harquahala and Buckskin Mountains, west central Arizona: Implications for denudation by a major detachment fault system, J. Geophys. Res., 95, 19,973-19,987, 1990.
Rosendahl, B. R., Architecture of continental rifts with special reference to East Africa, Annu. Rev. Earth Planet. Sci., 15, 445-503, 1987.
Sonder, L. J., C. H. Jones, and S. L. Salyards, Paleomagnetism of Miocene rocks near the Las Vegas Valley shear zone, Lake Mead region, southern Nevada: Spatially variable vertical axis rotations, Eos Trans. AGU, 70, 1070, 1989.
Spencer, J. E., Role of tectonic denudation in uplift and warping of low-angle normal faults, Geology, 12, 95-98, 1984.
Spencer, J. E., Miocene low-angle normal faulting and dike emplacement, Homer Mountain and surrounding areas, southeastern California and southernmost Nevada, Geol. Soc. Am. Bull., 96, 1140-1155, 1985.
Spencer, J. E., and S. J. Reynolds, Middle Tertiary tectonics of Arizona and adjacent areas, Ariz. Geol. Soc. Dig., 17, 539-574, 1989.
Steiger, R. H., and E. Jager, Subcommission on geochronology: Convention on the use of decay constants in geo- and cosmochronology, Earth Planet. Sci. Lett., 36, 359-362, 1977.
Thenhaus, P. C., and T. P. Barnhard, Regional termination and segmentation of Quaternary fault belts in the Great Basin, Nevada and Utah, Bull. Seismol. Soc. Am., 79, 1426-1438, 1989.
Weber, M. E., and E. I. Smith, Structural and geochemical constraints on the reassembly of disrupted mid-Miocene volcanoes in the Lake Mead-Eldorado Valley area of southern Nevada, Geology, 15, 553-556, 1987.
Wells, R. E., and J. W. Hillhouse, Paleomagnetism and tectonic rotation of the lower Miocene Peach Springs Tuff: Colorado Plateau, Arizona, to Barstow, California, Geol. Soc. Am. Bull., 101, 846-863, 1989.
Wernicke, B., Uniform-sense normal simple shear of the continental lithosphere, Can. J. Earth Sci., 22, 108-125, 1985.
Wernicke, B., and G. J. Axen, On the role of isostasy in the evolution of normal fault systems, Geology, 16, 848-851, 1988.
Wernicke, B., and B. C. Burchfiel, Modes of extensional tectonics, J. Struct. Geol., 4, 105-115, 1982.
Young, R. A., Paleogeomorphic evidence for the structural history of the Colorado Plateau margin in western Arizona, in Mesozoic-Cenozoic Tectonic Evolution of the Colorado River Region, California, Arizona, and Nevada, edited by E.G. Frost and D.L. Martin, pp. 29-40, Cordilleran, San Diego, Calif. 1982.

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